Comparative Study of Sampling-Based Path Planners

Marlon Garcia Honores
Department of Electrical Engineering
University of Central Florida
Email: ma724173@ucf.edu

I. Introduction

Nowadays, all mobile robots are require to be able to move from a starting point and arrive to goal points in order to do different tasks. However, it is could be difficult to move in a environment that is full of obstacles, since that, it is necessary to use path planning methods for the robots, in order to built a feasible path such that the robot would be able to avoid obstacles and reach the goal location. There are two main kind of methods, the grid based-methods and the sampling based methods.

The grid based methods divided the whole environment map in grids, as a way to discretize it, and try to find the best path by processing each grid. However, this require huge amount of memory and computational processing, that could mean a lot of time to find a path. Besides, if we move in a high dimensional space, this task may be impossible to solve for any CPU. On the other hand, the sample based methods offer a faster option, which consist in sample random points of the free feasible space in the map, and built a graph that would be increased with new point each iteration, in such a way that it looks as a tree with roots that are expanding and avoiding obstacles until we reach the desired goal location.

The present project will implement three classical sample methods in three different maps, these methods are Rapidly-exploring Random Tree (RRT), Optimal Rapidly-exploring Random Tree (RRT-start) and Rapidly-exploring Random Tree Connected (RRT-connect). Besides, as a novel contribution we will adapt the RRT with adaptive goal biasing, as a new method to show. Finally, we present the results for different parameters and we will make a comparison of the four methods.

II. Methods

A. RRT method

This method can be expressed in the Algorithm 1. Basically, as [1] explains, we will have a start point and we have to reach the goal point by constructing a tree. Each point of connection is labeled as node and the path from node to node as root or edge. The algorithm starts with the first node that is the start point and will random choose different points and check the nearest node to that point. After that, it will steer a root from the nearest node with a fixed step size on the direction of the random point, the final point using the fixed step could be the new node of the three as long as is free of collision with any obstacle. We keep doing this procedure until we reach the vicinity of the goal point or spent all the budget of nodes that we can save.

Algorithm 1 Rapidly-Exploring Random Tree (RRT)

Require: Initial point x_{start}, goal point x_{goal}, step size δ, node budget B
Ensure: A feasible path from x_{start} to x_{goal} (if found)

```
Initialize tree T ← {x_start}
while |T| < B do
    Sample a random point x_rand
    Find nearest node: x_near ← arg min_{x∈T} ||x − x_rand||
    Steer towards x_rand: x_new ← Steer(x_near, x_rand, δ)
    if the path from x_near to x_new is collision-free then
        Add node to tree: T ← T ∪ {x_new}
        Add edge (x_near, x_new)
        if ||x_new − x_goal|| < ϵ then
            return path from x_start to x_goal
        end if
    end if
end while
return Failure
```

B. RRT-star

RRT* is summarized in Algorithm 2. This method is similar to the standard RRT, with the key difference that each node is associated with a cost that must be optimized. As a consequence, the objective is to find a path with minimum cost, which can be interpreted as minimizing the traveled distance.

As mentioned in [2], at each iteration, a random sample is generated and a new node is added to the tree. Then, a local neighborhood is considered, and the algorithm evaluates whether the cost of reaching the new node can be reduced by selecting a different parent among nearby nodes. If a lower-cost connection is found, the tree is updated accordingly through a rewiring process.

Although RRT* is computationally more expensive than RRT, but it guarantees asymptotic optimality in terms of path cost.

C. RRT-connect

This method, as [3] claims, is quite similar to the classical RRT, but the difference is that there are two trees that can grow

Algorithm 2 RRT* (Rapidly-Exploring Random Tree Star)

Require: Initial point x_{start}, goal point x_{goal}, step size δ, node budget B, neighborhood radius r

Ensure: A (near-)optimal path from x_{start} to x_{goal} (if found)

Initialize tree $T \leftarrow \{x_{\text{start}}\}$
Initialize cost $c(x_{\text{start}}) \leftarrow 0$
while $|T| < B$ **do**
  Sample a random point x_{rand}
  Find nearest node: $x_{\text{near}} \leftarrow \arg\min_{x \in T} \|x - x_{\text{rand}}\|$
  Steer towards x_{rand}: $x_{\text{new}} \leftarrow \text{Steer}(x_{\text{near}}, x_{\text{rand}}, \delta)$
  if the path from x_{near} to x_{new} is collision-free **then**
    Find neighborhood: $X_{\text{near}} \leftarrow \{x \in T \mid \|x - x_{\text{new}}\| \leq r\}$
    Choose best parent:

$$x_{\text{parent}} \leftarrow \arg\min_{x \in X_{\text{near}}} [c(x) + \text{Cost}(x, x_{\text{new}})]$$

    Add node: $T \leftarrow T \cup \{x_{\text{new}}\}$
    Add edge $(x_{\text{parent}}, x_{\text{new}})$
    Update cost: $c(x_{\text{new}}) \leftarrow c(x_{\text{parent}}) + \text{Cost}(x_{\text{parent}}, x_{\text{new}})$
    for each $x \in X_{\text{near}}$ **do**
      if $c(x_{\text{new}}) + \text{Cost}(x_{\text{new}}, x) < c(x)$ **and** path is collision-free **then**
        Rewire tree: replace parent of x with x_{new}
        Update cost: $c(x) \leftarrow c(x_{\text{new}}) + \text{Cost}(x_{\text{new}}, x)$
      end if
    end for
    if $\|x_{\text{new}} - x_{\text{goal}}\| < \epsilon$ **then**
      Store candidate path to goal
    end if
  end if
end while
return best path found

from both sides , the goal point and the start point. The sample random points will be the same for both trees. The algorithm is finish when the two trees are joined, as Algorithm 3 shows.

D. RRT- Adaptive goal biasing

Normally when we do random sampling in the above methods we consider a goal biasing probability which is the probability that we use to sample the goal point, which may be very low. However, in this method we consider that this probability will be increased as long as we move close to the goal. The algorithm is explained in Algorithm 4, and we use an auxiliary uniform random variable q, in order to know if we sample the goal, as long as, it is less than the goal bias probability p_{goal}. In our experiments the goal bias update step is given by $\alpha = (p_{\max} - p_{\min})\frac{|T_k|}{B}$, where B is the node budget and $|T_k|$ is the number of nodes that already exist in the tree, if we move close to the goal we increment p_{goal} with α and if we move far from the goal, we decrease p_{goal} with α, as more nodes we have, more increment on α.

Algorithm 3 RRT-Connect

Require: Initial state x_{start}, goal state x_{goal}, step size δ, node budget B

Ensure: A feasible collision-free path from x_{start} to x_{goal} if found

Initialize start tree $T_a \leftarrow \{x_{\text{start}}\}$
Initialize goal tree $T_b \leftarrow \{x_{\text{goal}}\}$
while $|T_a| + |T_b| < B$ **do**
  Sample a random state x_{rand}
  $x_{\text{new}} \leftarrow \text{EXTEND}(T_a, x_{\text{rand}}, \delta)$
  if $x_{\text{new}} \neq$ Collision point **then**
    $x_{\text{connect}} \leftarrow \text{CONNECT}(T_b, x_{\text{new}}, \delta)$
    if $x_{\text{connect}} = x_{\text{new}}$ **then**
      return collision-free path obtained by joining T_a and T_b
    end if
  end if
  Swap trees $T_a \leftrightarrow T_b$
end while
return Failure

Algorithm 4 RRT with Adaptive Goal Biasing

Require: Initial point x_{start}, goal point x_{goal}, step size δ, node budget B

Require: Initial goal bias p_0, minimum bias $p_{\min}$, maximum bias $p_{\max}$ and bias update gain α.

Ensure: A feasible collision-free path from x_{start} to x_{goal} if found

Initialize tree $T \leftarrow \{x_{\text{start}}\}$, goal bias $p_{\text{goal}} \leftarrow p_0$, the best distance $d_{\text{best}} \leftarrow \|x_{\text{start}} - x_{\text{goal}}\|$
while $|T| < B$ **do**
  Sample $q \sim \mathcal{U}(0, 1)$
  if $q < p_{\text{goal}}$ **then**
    $x_{\text{rand}} \leftarrow x_{\text{goal}}$
  else
    $x_{\text{rand}} \leftarrow \text{RANDOM SAMPLE}()$
  end if
  $x_{\text{near}} \leftarrow \arg\min_{x \in T} \|x - x_{\text{rand}}\|$
  $x_{\text{new}} \leftarrow \text{STEER}(x_{\text{near}}, x_{\text{rand}}, \delta)$
  if $\text{COLLISIONFREE}(x_{\text{near}}, x_{\text{new}})$ **then**
    Add x_{new} to T and add edge $(x_{\text{near}}, x_{\text{new}})$
    $d_{\text{new}} \leftarrow \|x_{\text{new}} - x_{\text{goal}}\|$
    if $d_{\text{new}} < d_{\text{best}}$ **then**
      $d_{\text{best}} \leftarrow d_{\text{new}}$ and $p_{\text{goal}} \leftarrow \min(p_{\max}, p_{\text{goal}} + \alpha)$
    else
      $p_{\text{goal}} \leftarrow \max(p_{\min}, p_{\text{goal}} - \alpha)$
    end if
    if $d_{\text{new}} < \epsilon$ **then**
      return path from x_{start} to x_{goal}
    end if
  end if
end while
return $\varnothing$

TABLE I: Start and Goal Coordinates for Each Map (in pixel units)

Map	Start		Goal	
	x	y	x	y
Map A	25	25	475	475
Map B	250	250	750	750
Map C	500	500	2500	2500

TABLE II: Planning parameters

Parameter	Value
Goal radius (ϵ)	20
Obstacle threshold	100
Minimum goal bias	0.05
Maximum goal bias	0.60
Rewire radius (RRT-star)	50

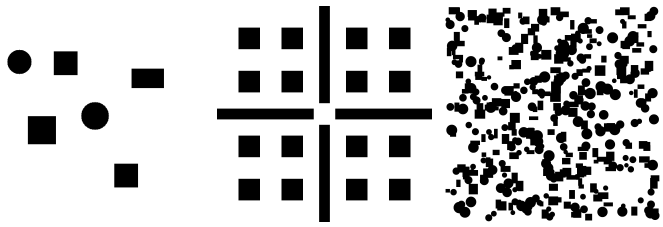
III. Experimental Setup

The parameters for each algorithm are be given in Table II, where the obstacle threshold in the distance that all algorithm can tolerate from the obstacle, the goal radius is the common stop condition for all algorithms, these measures are given in pixels units. The minimum and maximum goal bias are the probabilities corresponding to the RRT-adaptive bias, and the rewire radius is corresponding to the RRT-star. All the algorithms will present the same collision checker which is summarized in Algorithm 5.

Algorithm 5 Collision-Free Check Between Two Nodes

Require: Map $\mathcal{M}$, nodes $\mathbf{n}_1 = (x_1, y_1)$, $\mathbf{n}_2 = (x_2, y_2)$
Ensure: `True` if collision-free, otherwise `False`
$d \leftarrow \|\mathbf{n}_2 - \mathbf{n}_1\|$
$N \leftarrow \lfloor d \rfloor$, is the number of discrete points to check.
if $N = 0$ **then**
  if $\mathcal{M}(x_1, y_1) >$ Threshold **then**
    return True
  else
    return False
  end if
end if
for $i = 0$ to N **do**
  $t \leftarrow \frac{i}{N}$
  $x \leftarrow x_1 + t(x_2 - x_1)$
  $y \leftarrow y_1 + t(y_2 - y_1)$
  if $\mathcal{M}(\lfloor x \rfloor, \lfloor y \rfloor) \leq$ Threshold **then**
    return False
  end if
end for
return True

From the Algorithm 5, we discretize the distance between two nodes and we save it in N, and we play the function $\mathcal{M}(,)$ as value of the pixel in this point, if it is over certain threshold, it is an obstacle point and if it is below this threshold, would be a free collision point. In the algorithm we do interpolation for the intermediate points between the nodes, and check them, if there is an obstacle by using $\mathcal{M}(,)$.



(a) Map A (b) Map B (c) Map C

Fig. 1: Environment maps used in experiments

TABLE III: Evaluated parameter values

Parameter	Values
Node budget	$\{500,\ 1000,\ 3000,\ 5000\}$
Step size	$\{15,\ 30,\ 45,\ 60\}$

In order to do a comparison how the results change with different algorithm's parameters, we choose the step size of the roots and the node budgets as the parameters to vary, in that sense we use the values in Table III.

All the algorithms will be run 20 times for each map and for each specific value of the parameter listed in Table III, and the initial and goal points are given Table I. Since the algorithms are based in random sampling, each run will have different random seeds. Besides the maps that we are using are given in Figure 1.

At the moment that we try with different node budgets, we fix step size as 15, and at the moment that we change the step size we fix node budget as 5000. In the Section IV, the numerical results are presented as average value $\pm$ standard deviation for the 20 runs.

The results consist in time for getting the path, number of iterations, nodes used, success rate of finding path for the 20 runs and path length. Besides, graphs of the path along the maps are given in the Appendices with plots of the path length convergence along iterations for the case of RRT-star. In case we couldn't find the path, some data will appear as N/A, and the plots of convergence path length will not appear for RRT-star.

IV. Results

We present the maps with the paths according to the change of node budgets and step sizes, with the plots of convergence length path along the iterations for RRT-star, in the Appendices. Furthemore, the table of results with the different average measures from the 20 runs are given in the following sections with the chart bars for the success rate and path length.

A. *RRT results*

TABLE IV: RRT performance varying node budget (step size = 15)

Node Budget	Map	Time (s)	Iterations
500	A	0.006 ± 0.002	213.3 ± 60.6
500	B	0.016 ± 0.014	581.9 ± 387.8
500	C	N/A	N/A
1000	A	0.007 ± 0.002	213.3 ± 60.6
1000	B	0.032 ± 0.047	839.0 ± 837.5
1000	C	0.205 ± 0.000	3924.0 ± 0.0
3000	A	0.006 ± 0.002	213.3 ± 60.6
3000	B	0.039 ± 0.058	916.1 ± 882.7
3000	C	0.781 ± 0.391	5938.1 ± 1970.6
5000	A	0.006 ± 0.003	213.3 ± 60.6
5000	B	0.041 ± 0.060	916.1 ± 882.7
5000	C	0.956 ± 0.616	6450.1 ± 2435.9

TABLE V: RESULTS VARYING STEP SIZE (NODE BUDGET = 5000)

Step Size	Map	Time (s)	Iterations
30	A	0.004 ± 0.001	102.9 ± 24.0
30	B	0.017 ± 0.029	517.7 ± 613.4
30	C	0.322 ± 0.254	3875.1 ± 2215.9
45	A	0.012 ± 0.031	159.4 ± 270.6
45	B	0.123 ± 0.281	958.1 ± 1300.4
45	C	0.150 ± 0.097	2299.7 ± 1034.5
60	A	0.007 ± 0.011	112.8 ± 142.3
60	B	0.063 ± 0.143	761.7 ± 908.0
60	C	0.440 ± 0.799	2809.5 ± 2774.1

TABLE VI: RRT : PATH LENGTH AND NODES USED (NODE BUDGET VARIATION, STEP SIZE = 15)

Node Budget	Map	Path Length (px)	Nodes Used
500	A	783.25 ± 48.85	178.80 ± 49.54
500	B	968.24 ± 48.52	267.85 ± 119.15
500	C	N/A	500.00 ± 0.00
1000	A	783.25 ± 48.85	178.80 ± 49.54
1000	B	967.98 ± 47.75	308.60 ± 216.21
1000	C	3678.82 ± 0.00	994.05 ± 25.94
3000	A	783.25 ± 48.85	178.80 ± 49.54
3000	B	968.28 ± 46.56	317.90 ± 247.48
3000	C	4251.95 ± 505.62	1973.70 ± 659.22
5000	A	783.25 ± 48.85	178.80 ± 49.54
5000	B	968.28 ± 46.56	317.90 ± 247.48
5000	C	4235.17 ± 490.18	2127.50 ± 986.89

TABLE VII: RRT: PATH LENGTH AND NODES USED (STEP SIZE VARIATION, NODE BUDGET = 5000)

Step Size	Map	Path Length (px)	Nodes Used
30	A	796.52 ± 51.34	84.15 ± 16.67
30	B	989.44 ± 76.56	173.30 ± 154.26
30	C	4322.73 ± 656.23	1000.65 ± 474.38
45	A	793.37 ± 59.77	127.50 ± 218.22
45	B	998.73 ± 57.09	495.60 ± 760.89
45	C	4134.76 ± 476.80	888.35 ± 988.14
60	A	810.45 ± 64.87	83.05 ± 98.02
60	B	1016.96 ± 113.95	302.25 ± 462.99
60	C	4215.50 ± 457.84	1383.90 ± 1701.92

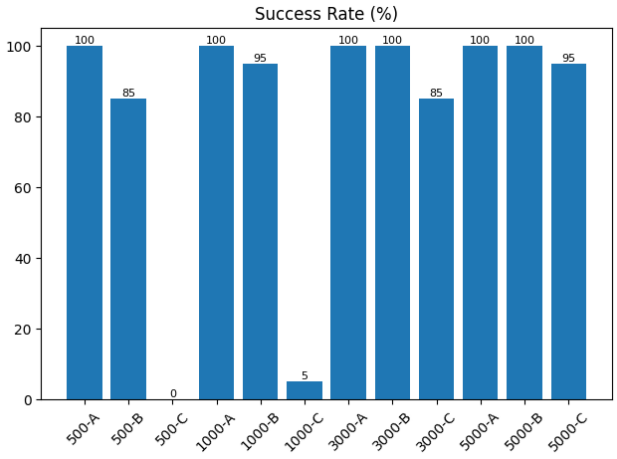


Fig. 2: Success Rate for different Maps and node budgets for RRT.

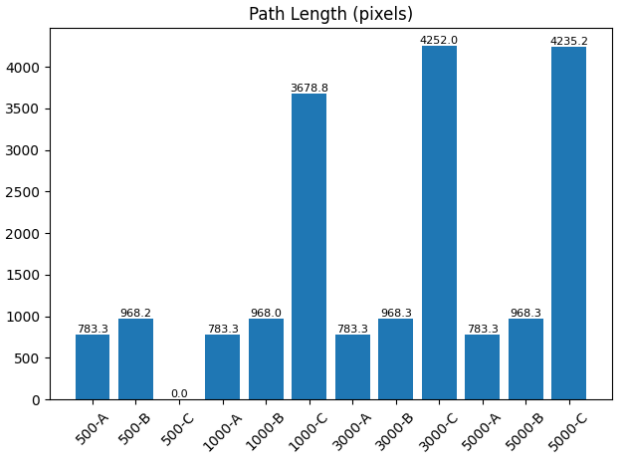


Fig. 3: Path length for different Maps and node budgets for RRT.

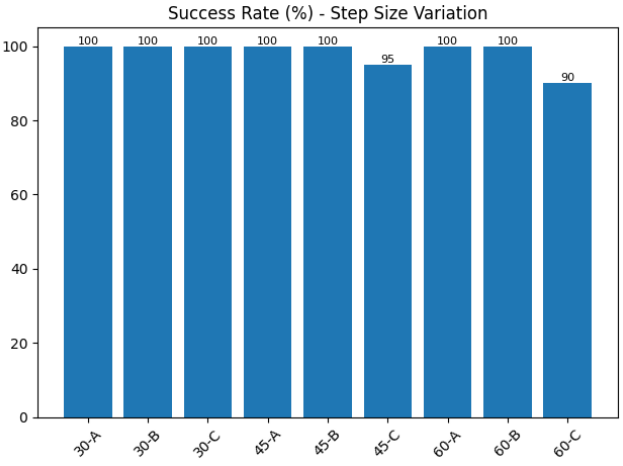


Fig. 4: Success Rate for different Maps and step-sizes for RRT.

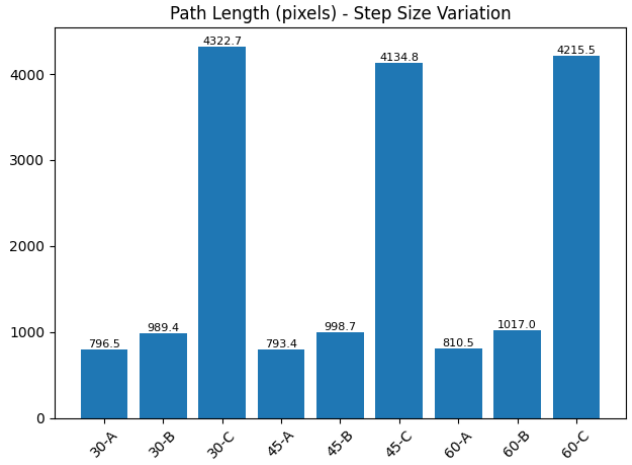


Fig. 5: Path length for different Maps and step-sizes for RRT.

B. *RRT-star results*

TABLE VIII: RRT* PERFORMANCE VARYING NODE BUDGET (STEP SIZE = 15)

Node Budget	Map	Time (s)	Iterations
500	A	0.211 ± 0.008	567.2 ± 18.3
500	B	0.193 ± 0.014	1106.7 ± 265.7
500	C	N/A	N/A
1000	A	0.786 ± 0.023	1121.8 ± 33.6
1000	B	0.625 ± 0.069	2165.9 ± 616.0
1000	C	0.457 ± 0.000	4204.0 ± 0.0
3000	A	5.954 ± 0.342	3419.3 ± 122.2
3000	B	3.469 ± 0.177	4972.1 ± 689.3
3000	C	2.593 ± 0.262	8293.6 ± 1801.8
5000	A	15.752 ± 0.765	5754.5 ± 202.4
5000	B	8.515 ± 0.251	7669.3 ± 705.9
5000	C	6.054 ± 0.338	11918.4 ± 1850.6

TABLE IX: RRT* PERFORMANCE VARYING STEP SIZE (NODE BUDGET = 5000)

Step Size	Map	Time (s)	Iterations
15	A	15.752 ± 0.765	5754.5 ± 202.4
15	B	8.515 ± 0.251	7669.3 ± 705.9
15	C	6.054 ± 0.338	11918.4 ± 1850.6
30	A	16.854 ± 1.517	6181.4 ± 234.8
30	B	10.133 ± 1.783	7880.7 ± 385.2
30	C	5.489 ± 0.189	10816.8 ± 1339.7
45	A	16.775 ± 0.829	6939.2 ± 347.9
45	B	10.846 ± 1.728	8384.8 ± 174.0
45	C	6.227 ± 1.006	10238.4 ± 949.1
60	A	16.987 ± 0.431	7609.9 ± 469.7
60	B	10.371 ± 1.621	9295.3 ± 299.8
60	C	6.262 ± 0.884	10491.0 ± 900.5

TABLE X: RRT* PATH LENGTH AND NODES USED (NODE BUDGET VARIATION, STEP SIZE = 15)

Node Budget	Map	Path Length (px)	Nodes Used
500	A	676.35 ± 28.58	500.00 ± 0.00
500	B	830.39 ± 37.62	500.00 ± 0.00
500	C	N/A	500.00 ± 0.00
1000	A	661.31 ± 7.47	1000.00 ± 0.00
1000	B	815.27 ± 30.36	1000.00 ± 0.00
1000	C	3306.19 ± 0.00	1000.00 ± 0.00
3000	A	647.13 ± 2.75	3000.00 ± 0.00
3000	B	793.87 ± 9.43	3000.00 ± 0.00
3000	C	3760.43 ± 424.13	3000.00 ± 0.00
5000	A	644.62 ± 1.79	5000.00 ± 0.00
5000	B	781.54 ± 6.45	5000.00 ± 0.00
5000	C	3739.86 ± 414.32	5000.00 ± 0.00

TABLE XI: RRT* PATH LENGTH AND NODES USED (STEP SIZE VARIATION, NODE BUDGET = 5000)

Step Size	Map	Path Length (px)	Nodes Used
15	A	644.62 ± 1.79	5000.00 ± 0.00
15	B	781.54 ± 6.45	5000.00 ± 0.00
15	C	3739.86 ± 414.32	5000.00 ± 0.00
30	A	644.74 ± 1.54	5000.00 ± 0.00
30	B	788.81 ± 7.72	5000.00 ± 0.00
30	C	4037.20 ± 618.10	5000.00 ± 0.00
45	A	645.87 ± 3.53	5000.00 ± 0.00
45	B	793.29 ± 14.37	5000.00 ± 0.00
45	C	4036.24 ± 480.90	5000.00 ± 0.00
60	A	645.72 ± 2.33	5000.00 ± 0.00
60	B	798.30 ± 11.25	5000.00 ± 0.00
60	C	4113.27 ± 455.34	5000.00 ± 0.00

TABLE XII: RRT* AVERAGE CONVERGENCE RATE PER 100 ITERATIONS (NODE BUDGET VARIATION, STEP SIZE = 15)

Node Budget	Map	Avg. Conv. Rate per 100 iterations
500	A	0.618 ± 1.655
500	B	0.448 ± 0.834
500	C	N/A
1000	A	2.127 ± 3.054
1000	B	0.927 ± 1.425
1000	C	1.002 ± 0.000
3000	A	1.147 ± 0.948
3000	B	0.765 ± 0.969
3000	C	0.106 ± 0.372
5000	A	0.722 ± 0.550
5000	B	0.755 ± 0.612
5000	C	0.015 ± 0.044

TABLE XIII: RRT* AVERAGE CONVERGENCE RATE PER 100 ITERATIONS (STEP SIZE VARIATION, NODE BUDGET = 5000)

Step Size	Map	Avg. Conv. Rate per 100 iterations
15	A	0.722 ± 0.550
15	B	0.755 ± 0.612
15	C	0.015 ± 0.044
30	A	1.621 ± 0.542
30	B	2.092 ± 1.143
30	C	0.015 ± 0.043
45	A	1.942 ± 0.874
45	B	2.284 ± 0.799
45	C	0.081 ± 0.164
60	A	2.299 ± 1.008
60	B	2.237 ± 0.975
60	C	0.015 ± 0.062

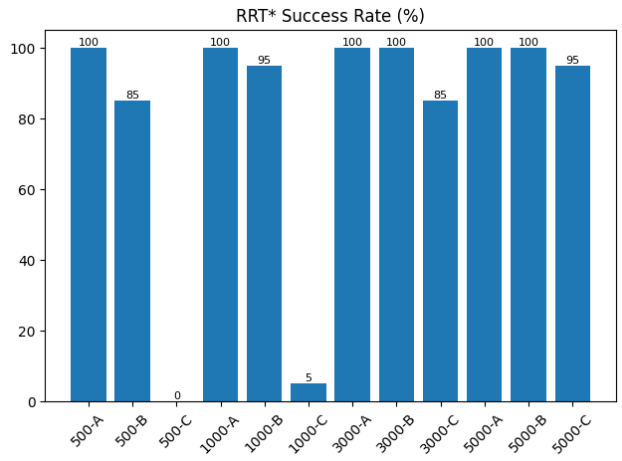


Fig. 6: Success Rate for different Maps and node budgets for RRT-star.

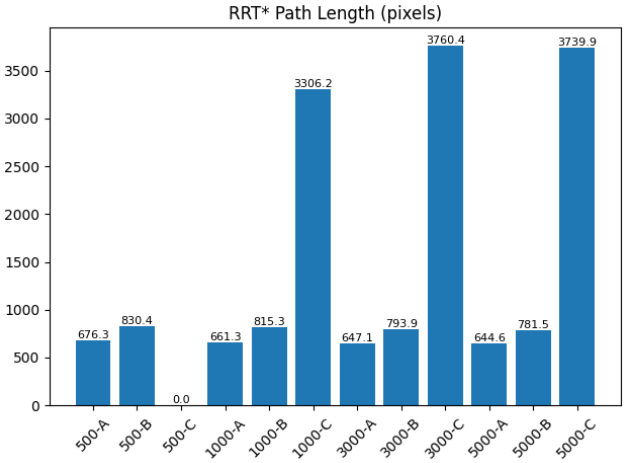


Fig. 7: Path length for different Maps and node budgets for RRT-star.

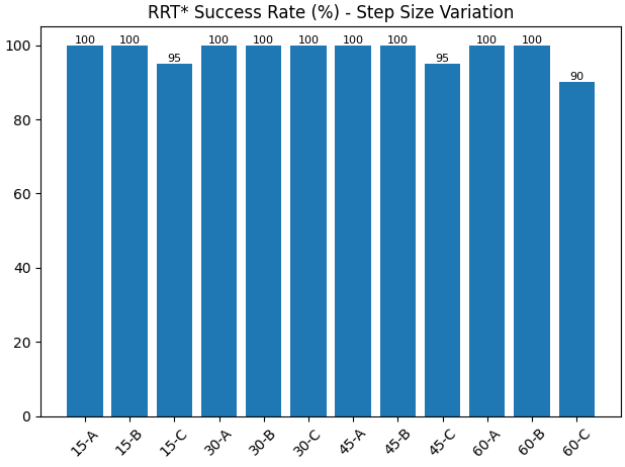


Fig. 8: Success Rate for different Maps and step-sizes for RRT-star.

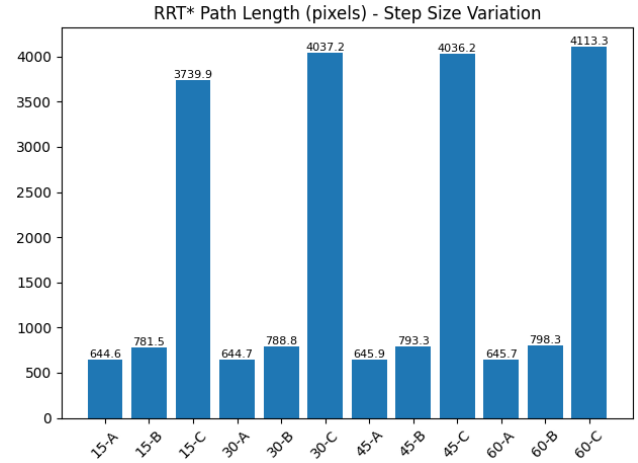


Fig. 9: Path length for different Maps and step-sizes for RRT-star.

C. *RRT-connect results*

TABLE XIV: RRT-CONNECT PERFORMANCE VARYING NODE BUDGET (STEP SIZE = 15)

Node Budget	Map	Time (s)	Iterations
500	A	0.001 ± 0.001	28.4 ± 8.7
500	B	0.003 ± 0.002	112.0 ± 68.7
500	C	0.029 ± 0.011	656.3 ± 225.3
1000	A	0.002 ± 0.001	28.4 ± 8.7
1000	B	0.003 ± 0.001	112.0 ± 68.7
1000	C	0.061 ± 0.049	1204.2 ± 818.0
3000	A	0.002 ± 0.001	28.4 ± 8.7
3000	B	0.003 ± 0.001	112.0 ± 68.7
3000	C	0.100 ± 0.097	1773.0 ± 1381.7
5000	A	0.002 ± 0.000	28.4 ± 8.7
5000	B	0.003 ± 0.001	112.0 ± 68.7
5000	C	0.103 ± 0.101	1773.0 ± 1381.7

TABLE XV: RRT-CONNECT PERFORMANCE VARYING STEP SIZE (NODE BUDGET = 5000)

Step Size	Map	Time (s)	Iterations
15	A	0.002 ± 0.000	28.4 ± 8.7
15	B	0.003 ± 0.001	112.0 ± 68.7
15	C	0.103 ± 0.101	1773.0 ± 1381.7
30	A	0.001 ± 0.000	17.3 ± 10.0
30	B	0.002 ± 0.001	48.4 ± 16.9
30	C	0.041 ± 0.039	977.1 ± 882.5
45	A	0.001 ± 0.000	12.1 ± 6.1
45	B	0.002 ± 0.001	64.1 ± 64.4
45	C	0.045 ± 0.047	1073.2 ± 856.0
60	A	0.001 ± 0.001	9.2 ± 4.5
60	B	0.003 ± 0.002	76.2 ± 80.1
60	C	0.029 ± 0.024	767.3 ± 666.9

TABLE XVI: RRT-CONNECT PATH LENGTH AND NODES USED (NODE BUDGET VARIATION, STEP SIZE = 15)

Node Budget	Map	Path Length (px)	Nodes Used
500	A	716.76 ± 25.40	62.25 ± 5.72
500	B	927.35 ± 51.31	105.80 ± 24.32
500	C	3383.39 ± 120.88	471.70 ± 57.17
1000	A	716.76 ± 25.40	62.25 ± 5.72
1000	B	927.35 ± 51.31	105.80 ± 24.32
1000	C	3608.53 ± 369.07	654.50 ± 238.35
3000	A	716.76 ± 25.40	62.25 ± 5.72
3000	B	927.35 ± 51.31	105.80 ± 24.32
3000	C	3632.67 ± 400.46	693.30 ± 305.39
5000	A	716.76 ± 25.40	62.25 ± 5.72
5000	B	927.35 ± 51.31	105.80 ± 24.32
5000	C	3632.67 ± 400.46	693.30 ± 305.39

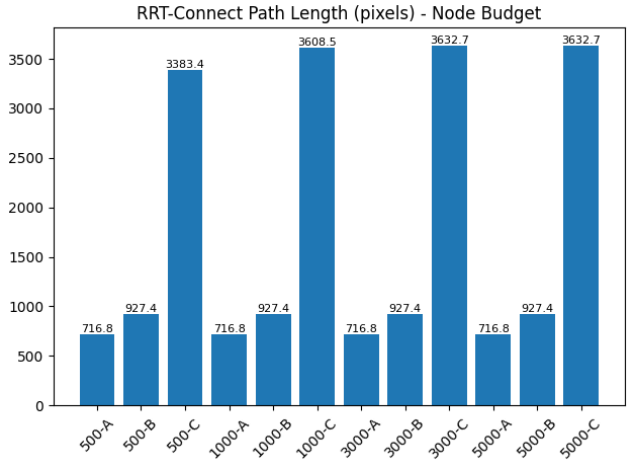


Fig. 11: Path length for different Maps and node budgets for RRT-connect.

TABLE XVII: RRT-CONNECT PATH LENGTH AND NODES USED (STEP SIZE VARIATION, NODE BUDGET = 5000)

Step Size	Map	Path Length (px)	Nodes Used
15	A	716.76 ± 25.40	62.25 ± 5.72
15	B	927.35 ± 51.31	105.80 ± 24.32
15	C	3632.67 ± 400.46	693.30 ± 305.39
30	A	728.14 ± 36.40	32.15 ± 5.29
30	B	932.20 ± 70.25	49.65 ± 8.08
30	C	3547.37 ± 256.85	356.25 ± 202.85
45	A	748.86 ± 46.83	22.85 ± 3.20
45	B	959.31 ± 86.21	39.70 ± 12.23
45	C	3883.76 ± 494.49	318.60 ± 201.31
60	A	756.07 ± 53.69	18.10 ± 3.02
60	B	965.12 ± 107.03	35.85 ± 17.64
60	C	3674.98 ± 472.31	215.20 ± 119.25

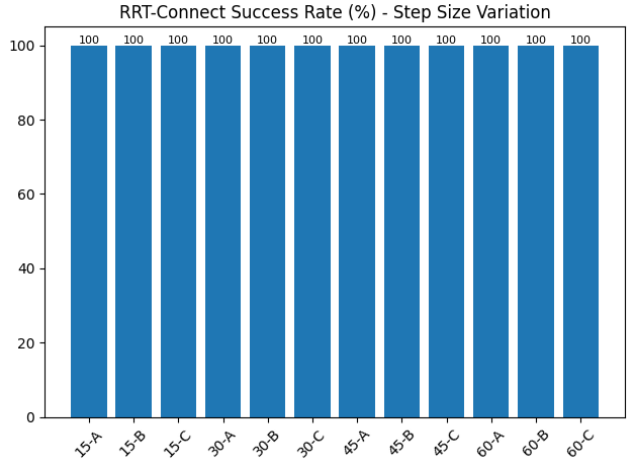


Fig. 12: Success Rate for different Maps and step-sizes for RRT-connect.

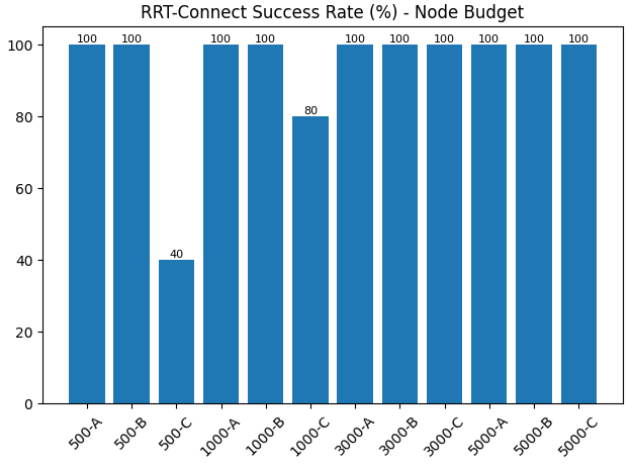


Fig. 10: Success Rate for different Maps and node budgets for RRT-connect.

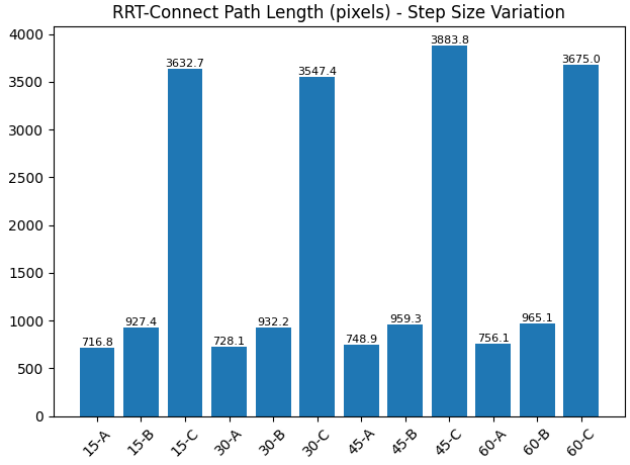


Fig. 13: Path length for different Maps and step-sizes for RRT-connect.

TABLE XX: RRT ADAPTIVE BIAS PATH LENGTH AND NODES USED (NODE BUDGET VARIATION, STEP SIZE = 15)

Node Budget	Map	Path Length (px)	Nodes Used
500	A	764.45 ± 34.24	139.65 ± 25.02
500	B	951.28 ± 49.12	215.15 ± 80.02
500	C	N/A	500.00 ± 0.00
1000	A	768.16 ± 35.21	157.30 ± 32.81
1000	B	956.38 ± 46.73	362.60 ± 293.38
1000	C	3968.30 ± 369.33	992.75 ± 19.19
3000	A	765.18 ± 26.12	181.50 ± 27.71
3000	B	981.08 ± 77.14	276.15 ± 172.24
3000	C	4123.65 ± 564.53	1770.35 ± 815.48
5000	A	772.24 ± 30.24	199.90 ± 35.93
5000	B	970.25 ± 58.72	322.60 ± 338.26
5000	C	4113.56 ± 369.01	1967.30 ± 860.03

TABLE XXI: RRT ADAPTIVE BIAS PATH LENGTH AND NODES USED (STEP SIZE VARIATION, NODE BUDGET = 5000)

Step Size	Map	Path Length (px)	Nodes Used
15	A	772.24 ± 30.24	199.90 ± 35.93
15	B	970.25 ± 58.72	322.60 ± 338.26
15	C	4113.56 ± 369.01	1967.30 ± 860.03
30	A	821.92 ± 42.20	106.80 ± 23.08
30	B	971.14 ± 65.46	209.00 ± 204.72
30	C	4064.02 ± 594.68	817.50 ± 667.50
45	A	809.70 ± 67.39	135.95 ± 206.47
45	B	978.31 ± 62.35	138.95 ± 124.81
45	C	4321.11 ± 600.80	1216.70 ± 1529.84
60	A	837.20 ± 73.90	215.10 ± 349.61
60	B	1081.59 ± 142.06	496.95 ± 1069.32
60	C	4181.40 ± 433.34	1480.75 ± 1805.76

D. RRT-adaptive bias results

TABLE XVIII: RRT ADAPTIVE GOAL BIAS VARYING NODE BUDGET (STEP SIZE = 15)

Node Budget	Map	Time (s)	Iterations
500	A	0.004 ± 0.001	169.6 ± 35.0
500	B	0.013 ± 0.010	523.4 ± 314.8
500	C	N/A	N/A
1000	A	0.005 ± 0.002	188.3 ± 38.2
1000	B	0.021 ± 0.025	623.8 ± 463.1
1000	C	0.356 ± 0.070	4488.0 ± 686.7
3000	A	0.006 ± 0.001	213.4 ± 32.5
3000	B	0.025 ± 0.032	624.2 ± 430.6
3000	C	0.748 ± 0.656	5319.1 ± 3094.6
5000	A	0.007 ± 0.002	235.5 ± 44.7
5000	B	0.040 ± 0.098	748.7 ± 806.1
5000	C	1.044 ± 0.936	6212.3 ± 3376.4

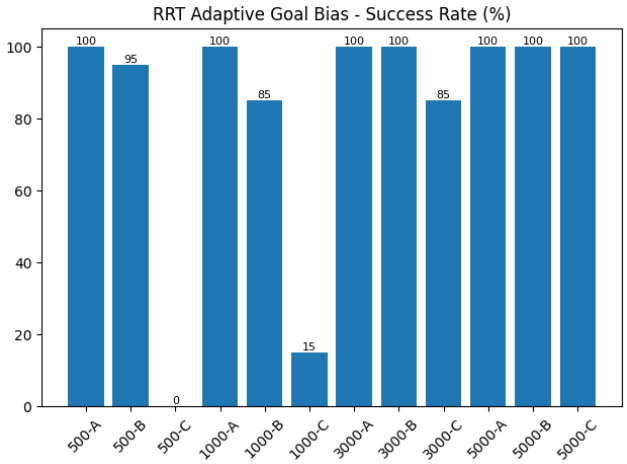


Fig. 14: Success Rate for different Maps and node budgets for RRT-adaptive goal bias.

TABLE XIX: RRT ADAPTIVE GOAL BIAS VARYING STEP SIZE (NODE BUDGET = 5000)

Step Size	Map	Time (s)	Iterations
15	A	0.007 ± 0.002	235.5 ± 44.7
15	B	0.040 ± 0.098	748.7 ± 806.1
15	C	1.044 ± 0.936	6212.3 ± 3376.4
30	A	0.005 ± 0.001	127.5 ± 27.7
30	B	0.021 ± 0.032	526.7 ± 452.8
30	C	0.292 ± 0.604	2867.0 ± 2900.8
45	A	0.012 ± 0.027	172.2 ± 268.5
45	B	0.013 ± 0.017	352.7 ± 329.5
45	C	0.275 ± 0.556	2807.8 ± 2450.7
60	A	0.032 ± 0.063	296.3 ± 482.3
60	B	0.039 ± 0.056	638.1 ± 622.7
60	C	0.145 ± 0.161	2067.8 ± 1017.7

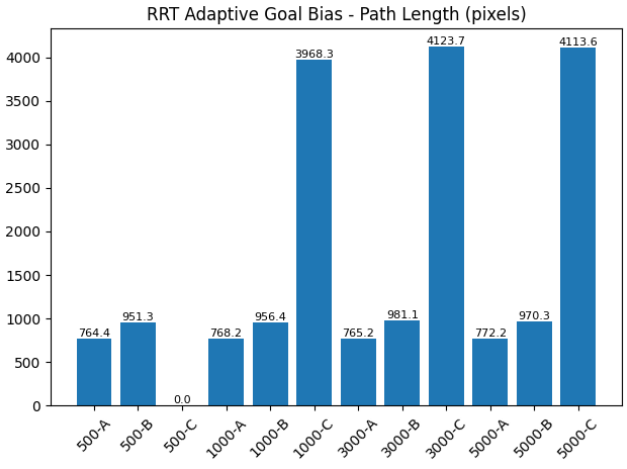


Fig. 15: Path length for different Maps and node budgets for RRT-adaptive goal bias.

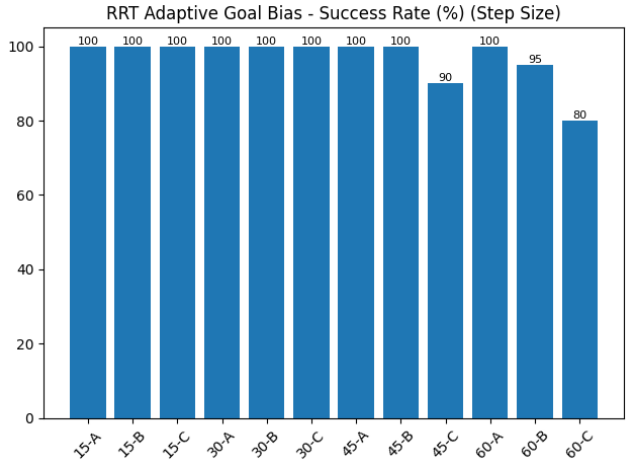


Fig. 16: Success Rate for different Maps and step-sizes for RRT-adaptive goal bias.

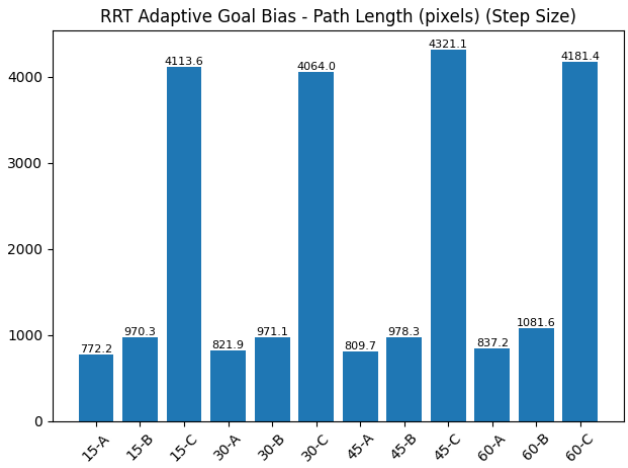


Fig. 17: Path length for different Maps and step-sizes for RRT-adaptive goal bias.

V. Discussion

As we can see on the presented results, all the methods arrived to the goal point with different step sizes and maps, however, for different budgets, the map C presented a huge difficult when the node budget is 500, in such a way that the success rate is 0 percent for RRT, RRT-star and RRT-adaptive goal bias methods, which means that node budget is the most important factor at the moment to ensure reaching the goal point. Besides, the RRT-connect method also present a low success rate (40 percent) with this map. We can see the paths that never reach goal in Appendices, belong to this node budget. This issue can be reformulated as a constrained optimization problem on budget, and there other approaches that can help to solve this as [4] presents.

Since the map C is the most difficult map, it might be the standard for comparison. For RRT, the best step size with higher succes rate is 45, however the path length is one of the highest, as well. The same result is showed for RRT-star and RRT adaptive goal bias. Otherwise, the RRT-connect method shows 100 success rate for all step sizes. Furthermore, we can claim that the best node budget is 5000 for all methods in terms of success rate, but as consequence, the methods tend to use more nodes and higher path lengths with this budget.

For the map C, the method with best performance is the RRT-connect with 15 step size and a budget of 5000 nodes in terms of success rate and path length. The main reason is that this algorithm considers always the position of the goal and construct a tree from this location, as a consequence, it tends to find a path faster. This method could be useful for high-dimensional spaces and applied to complex robots as [5] and [6] do.

For the results of RRT-star, we can see that the convergence of path length experimented notorious changes when we run the algorithm on Map A and B, but not for Map C, because in Map C, it is quite difficult to find new optimal nodes when the environment is full of obstacles. Furthermore, RRT-star produce shorter paths than RRT but invest more nodes, because its algorithm's structure aims to find optimal nodes that save path length, but it requires to spend more iterations to find those nodes in such a way that spends all the node budget.

The proposed method RRT-adaptive goal bias fall short with the node budget of 500, it presents a 0 percent of success rate. Besides, it presents greater path length than RRT-connect and RRT-star but less than the classical RRT. Sample the goal is not quite effective as start making a tree from there.

The practioner must consider select a high node budget in order to be able to complete complex maps. After that, the selection of the algorithm should come with the specific requirements and conditions. If the designer is able to know the goal location RRT-connect is the best option, but if it is not able to know it, unless the tree is close, the other methods are feasible. If the designer want to optimize path length as a cost to minimize, the RRT- star is the best option.

VI. Conclusion

In this project, we are able to show that RRT-connect method is the most effective for complex maps in terms of constrained node budget, however its success rate is low. Therefore, it is necessary to modify the methods in order to adjust its functionality for constrained node budget. Furthermore, RRT-star and RRT-connect present two different approaches that the designer could take advantage of them according to the requirements and the accessibility of the goal location.

Finally, the RRT-adaptive goal bias is presented as an optimal method that is less effective than RRT-connect but is able to have access to the goal with a controlled probability, which is a feature that could be more flexible.

Appendix

A. *RRT results*

Paths with different node budgets:

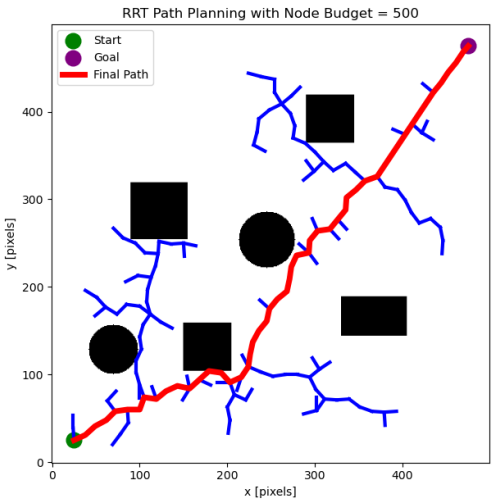


Fig. 18: Map A with RRT with 500 of node budget.

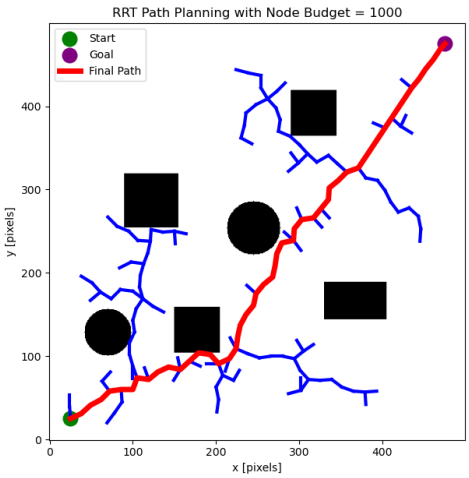


Fig. 21: Map A with RRT with 1000 of node budget.

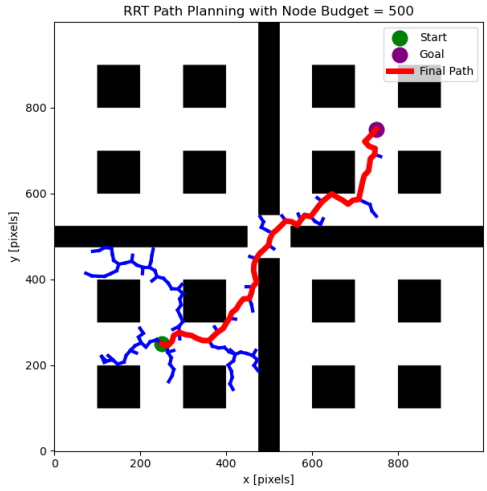


Fig. 19: Map B with RRT with 500 of node budget.

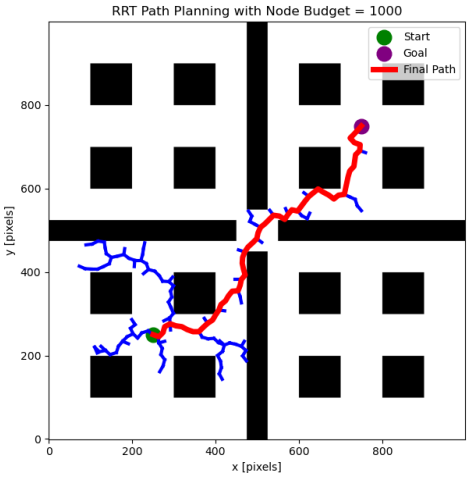


Fig. 22: Map B with RRT with 1000 of node budget.

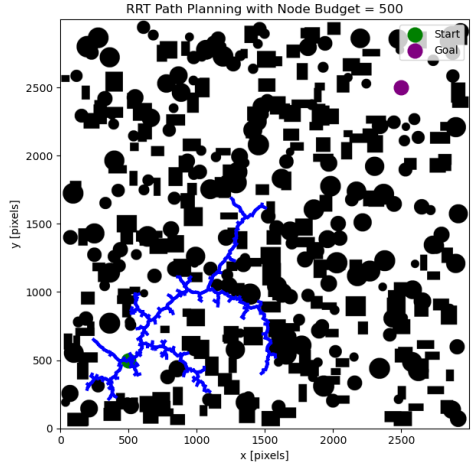


Fig. 20: Map C with RRT with 500 of node budget.

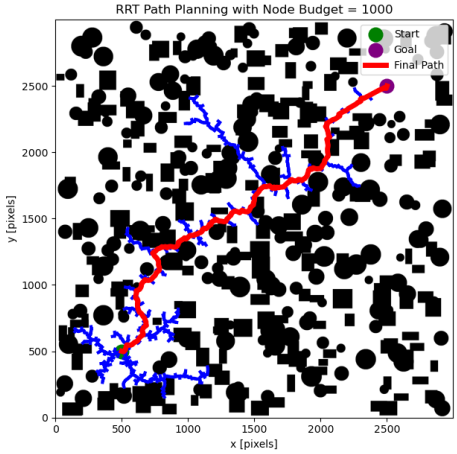


Fig. 23: Map C with RRT with 1000 of node budget.

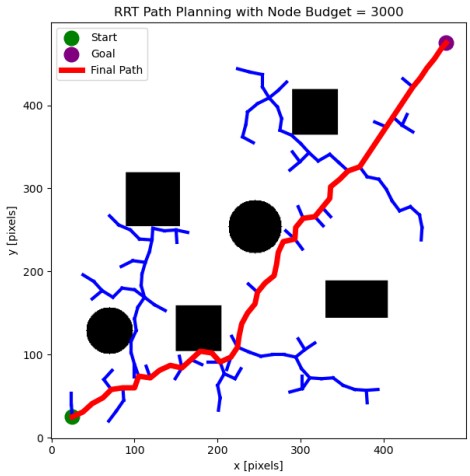


Fig. 24: Map A with RRT with 3000 of node budget.

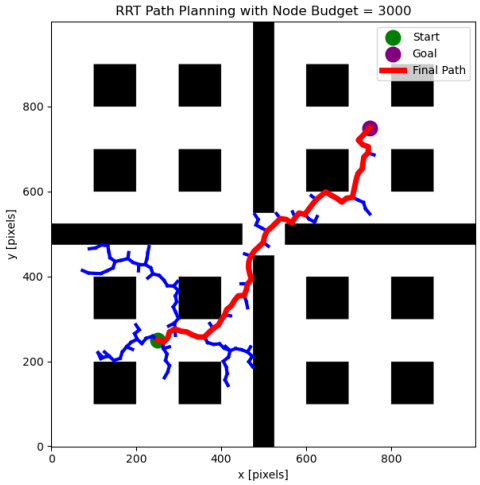


Fig. 25: Map B with RRT with 3000 of node budget.

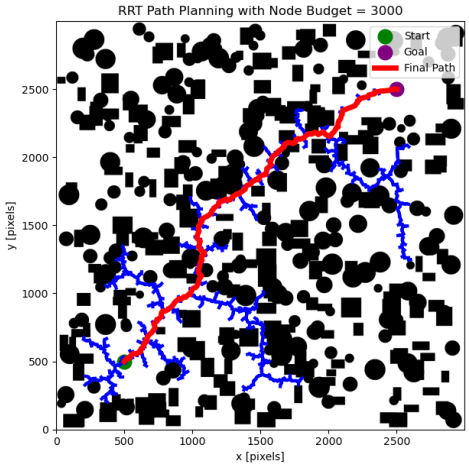


Fig. 26: Map C with RRT with 3000 of node budget.

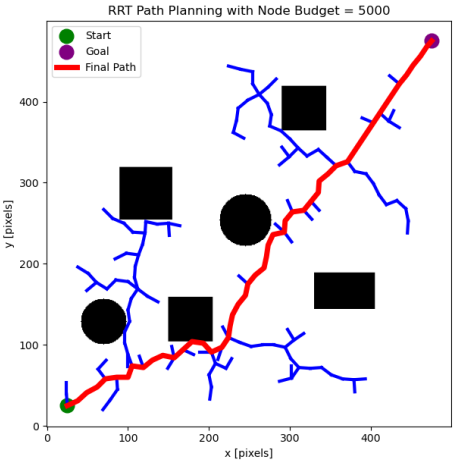


Fig. 27: Map A with RRT with 5000 of node budget.

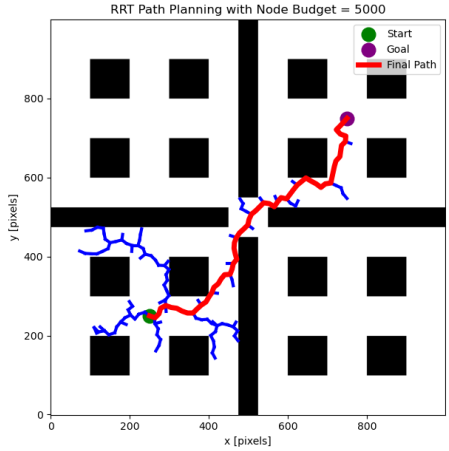


Fig. 28: Map B with RRT with 5000 of node budget.

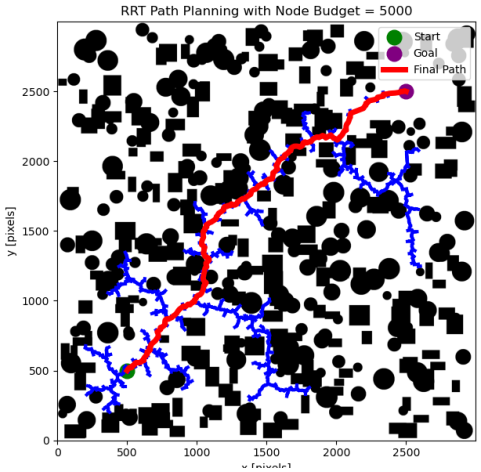


Fig. 29: Map C with RRT with 5000 of node budget.

For the different step-sizes , we consider the ones with step size 15 are already given in the Figures 27-29, so we have:

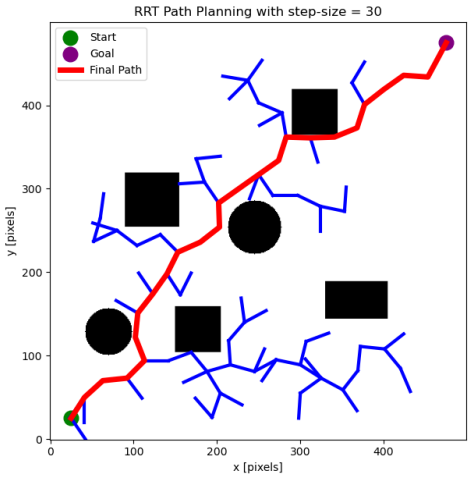


Fig. 30: Map A with RRT with 30 of step-size.

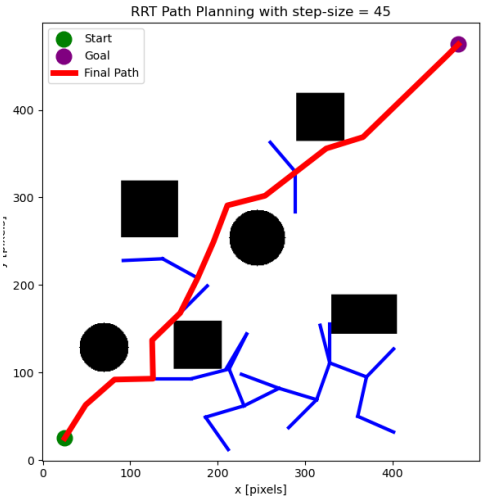


Fig. 33: Map A with RRT with 45 of step-size.

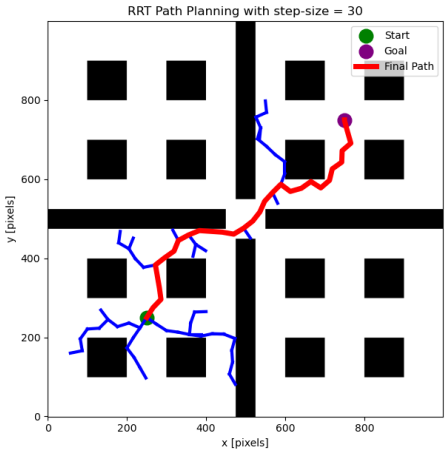


Fig. 31: Map B with RRT with 30 of step-size.

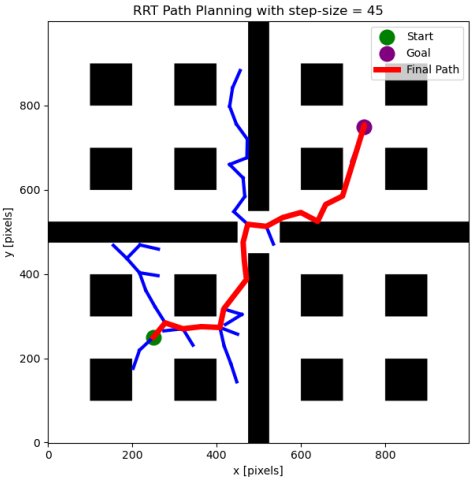


Fig. 34: Map B with RRT with 45 of step-size.

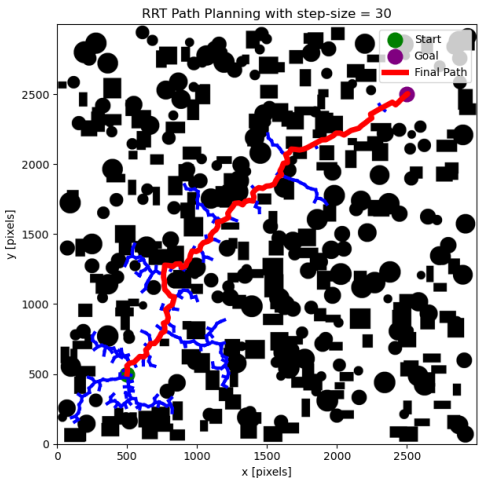


Fig. 32: Map C with RRT with 30 of step size.

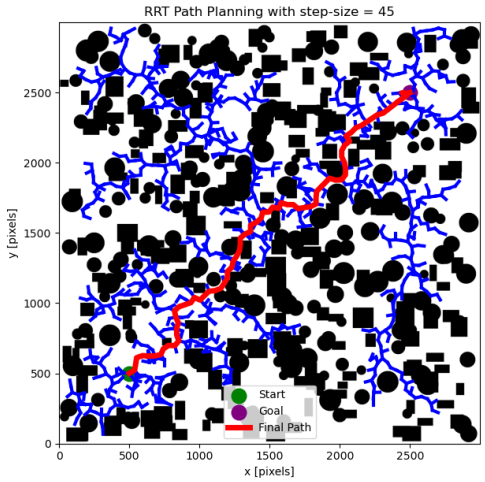


Fig. 35: Map C with RRT with 45 of step size.

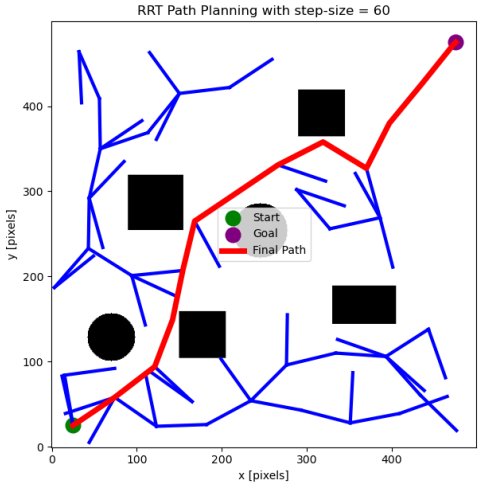


Fig. 36: Map A with RRT with 60 of step-size.

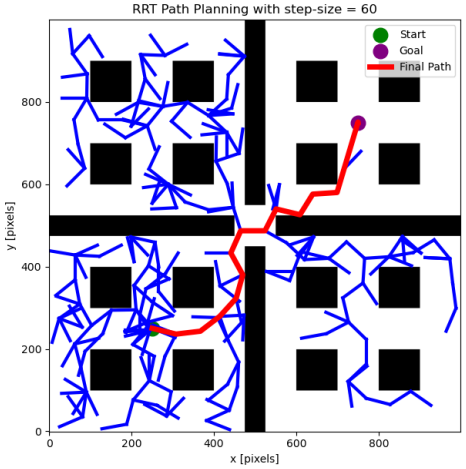


Fig. 37: Map B with RRT with 60 of step-size.

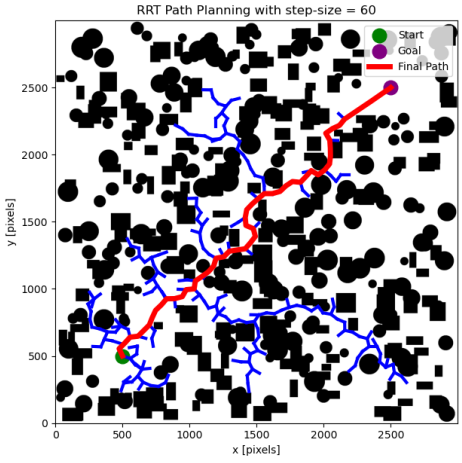


Fig. 38: Map C with RRT with 60 of step size.

B. *RRT-star results*

Paths with different node budget:

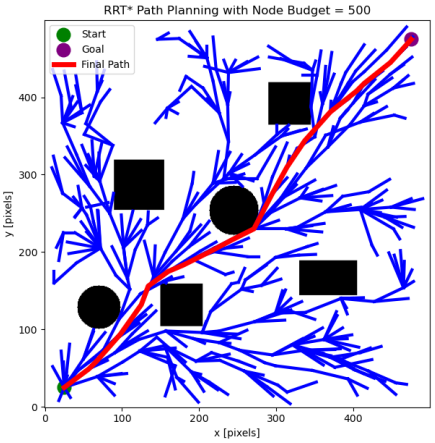


Fig. 39: Map A with RRT-start with a node budget of 500.

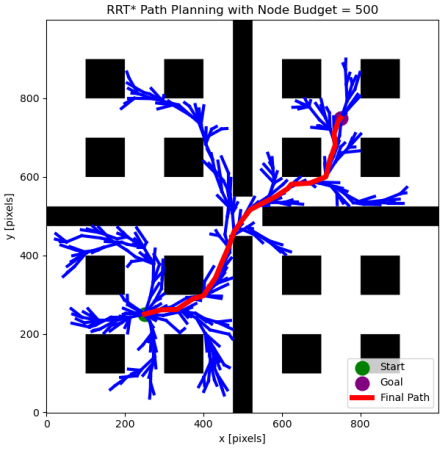


Fig. 40: Map B with RRT-start with a node budget of 500.

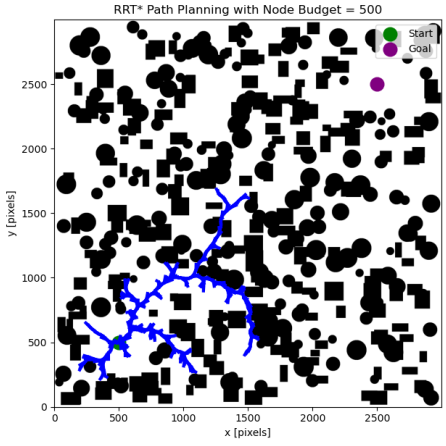


Fig. 41: Map C with RRT-start with a node budget of 500.

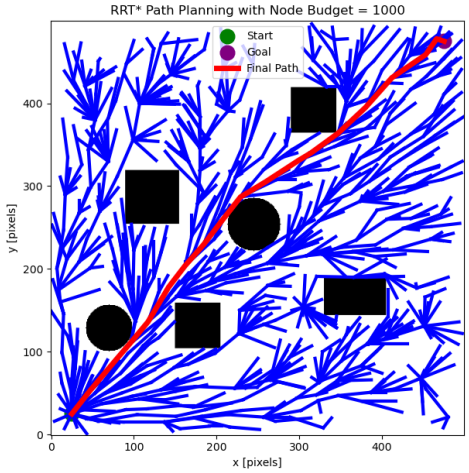


Fig. 42: Map A with RRT-start with a node budget of 1000.

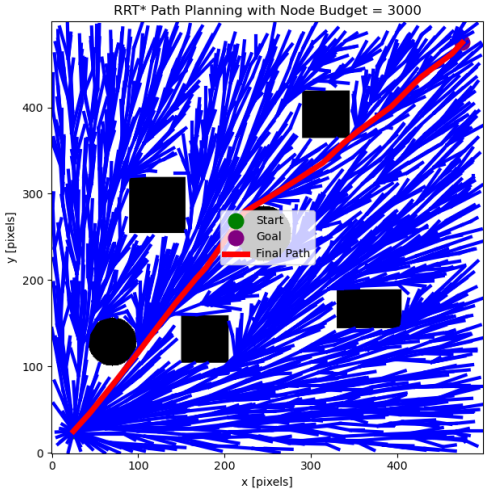


Fig. 45: Map A with RRT-start with a node budget of 3000.

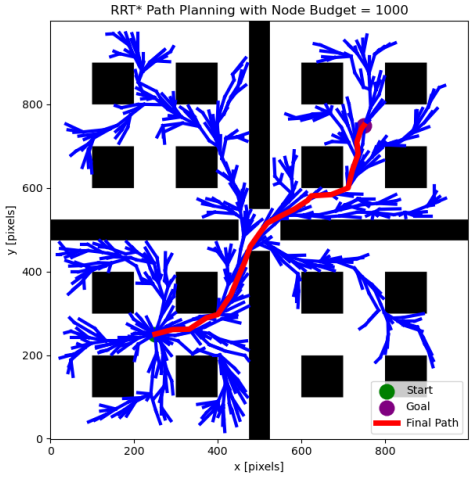


Fig. 43: Map B with RRT-start with a node budget of 1000.

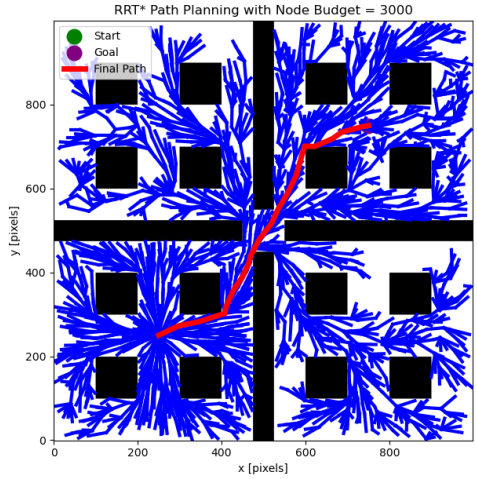


Fig. 46: Map B with RRT-start with a node budget of 3000.

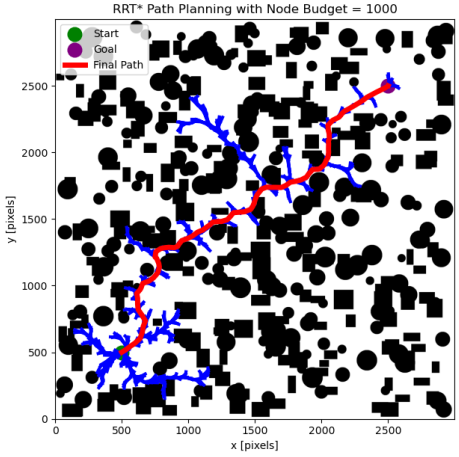


Fig. 44: Map C with RRT-start with a node budget of 1000.

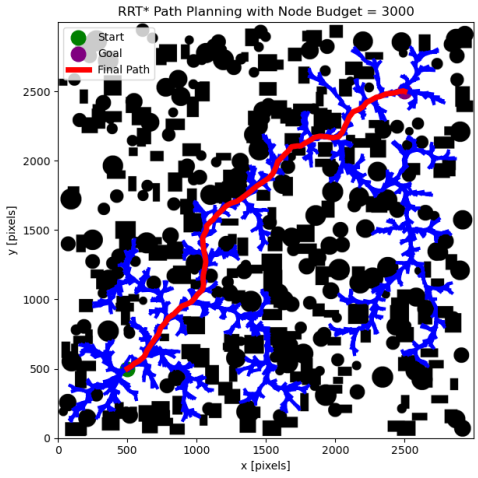


Fig. 47: Map C with RRT-start with a node budget of 3000.

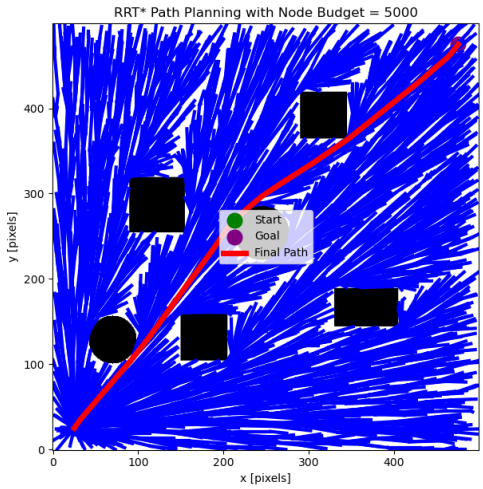


Fig. 48: Map A with RRT-start with a node budget of 5000.

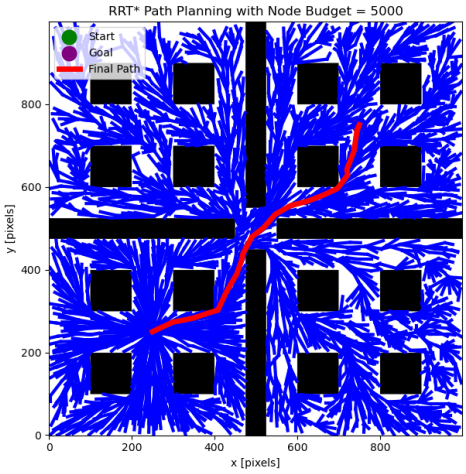


Fig. 49: Map B with RRT-start with a node budget of 5000.

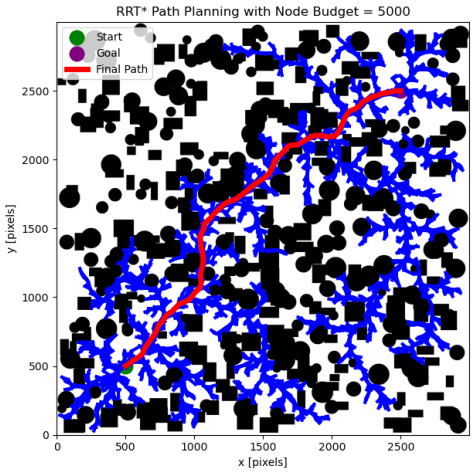


Fig. 50: Map C with RRT-start with a node budget of 5000.

For the different step-sizes , we consider the ones with step size 15 are already given in the Figures 48-50, so we have:

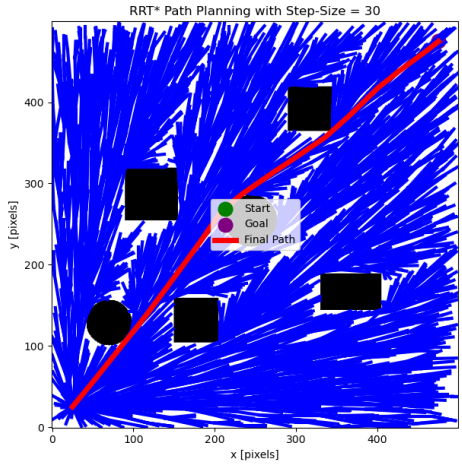


Fig. 51: Map A with RRT-start with a step size of 30.

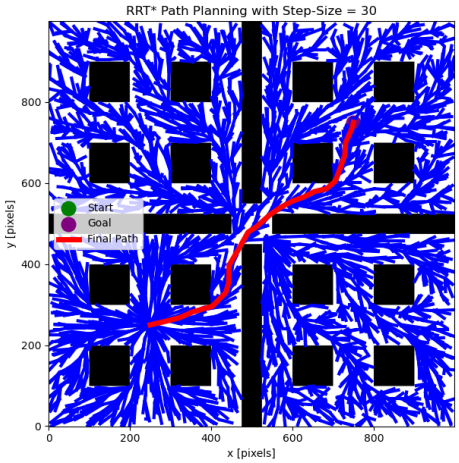


Fig. 52: Map B with RRT-start with a step size of 30.

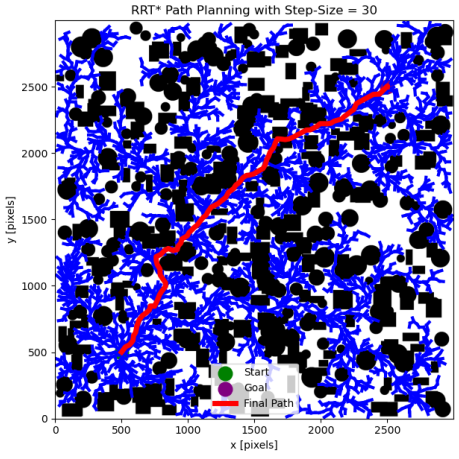


Fig. 53: Map C with RRT-start with a step size of 30.

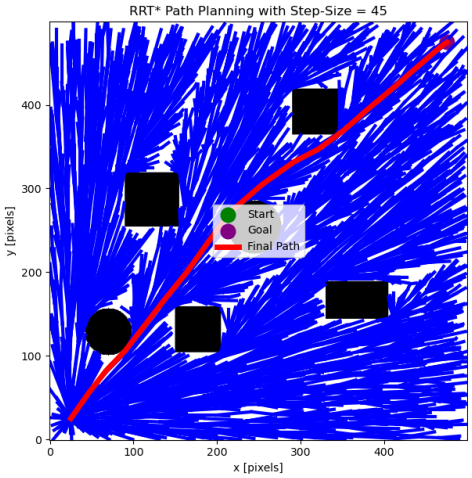


Fig. 54: Map A with RRT-start with a step size of 45.

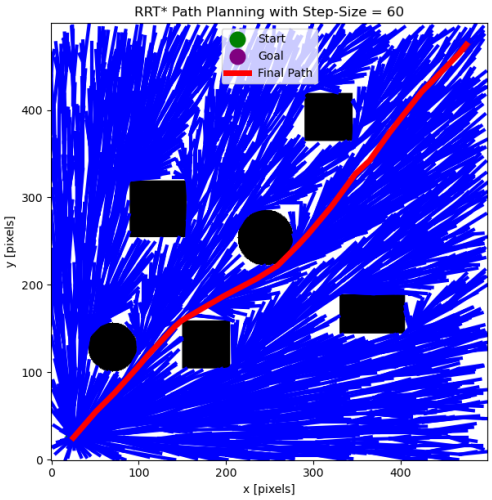


Fig. 57: Map A with RRT-start with a step size of 60.

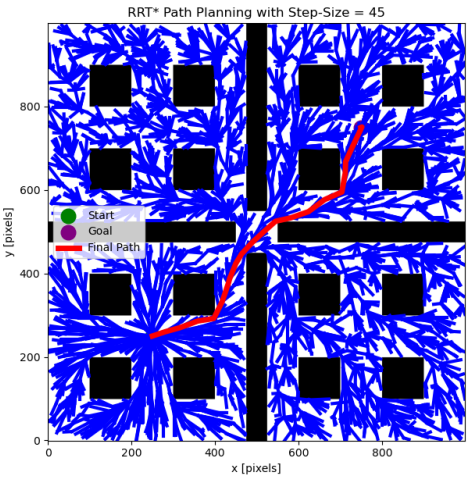


Fig. 55: Map B with RRT-start with a step size of 45.

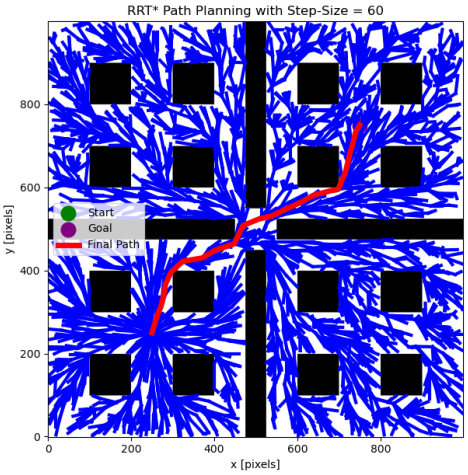


Fig. 58: Map B with RRT-start with a step size of 60.

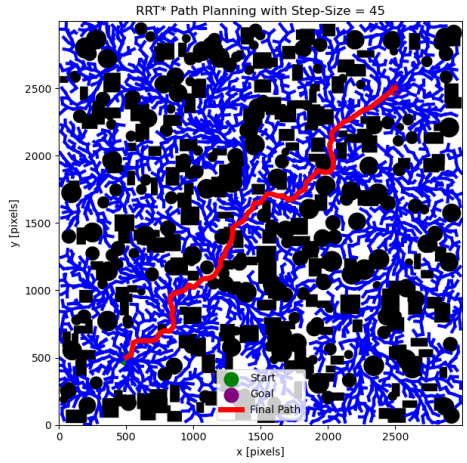


Fig. 56: Map C with RRT-start with a step size of 45.

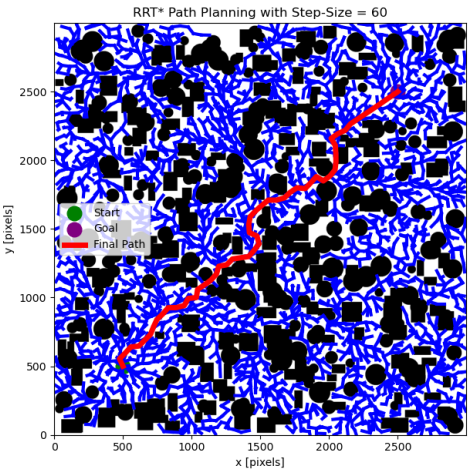


Fig. 59: Map C with RRT-start with a step size of 60.

Evolution of the path length along the iterations for different node budgets:

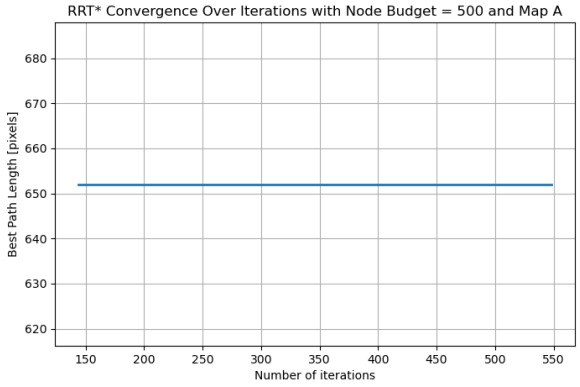


Fig. 60: Convergence of length path along the iterations for Map A, node budget of 500.

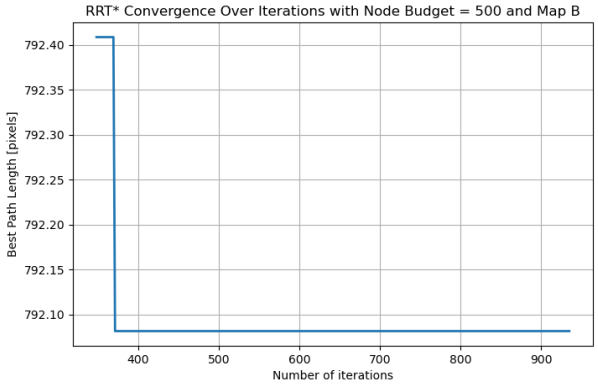


Fig. 61: Convergence of length path along the iterations for Map B, node budget of 500.

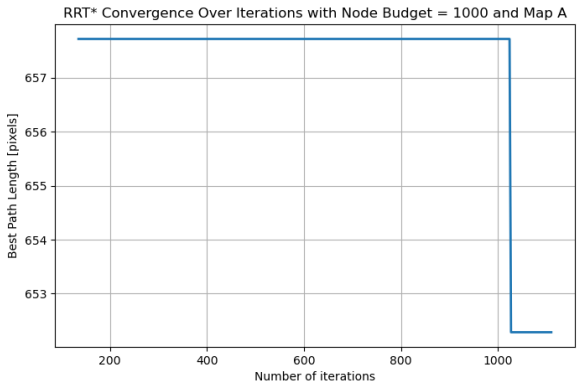


Fig. 62: Convergence of length path along the iterations for Map A, node budget of 1000.

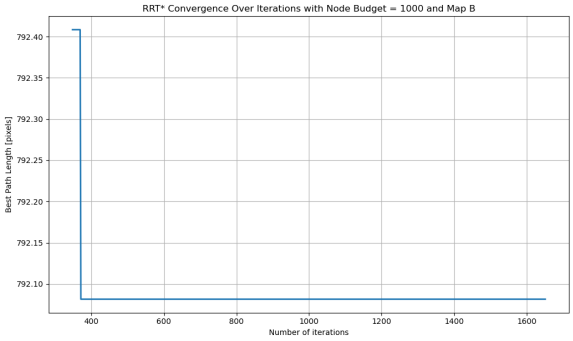


Fig. 63: Convergence of length path along the iterations for Map B, node budget of 1000.

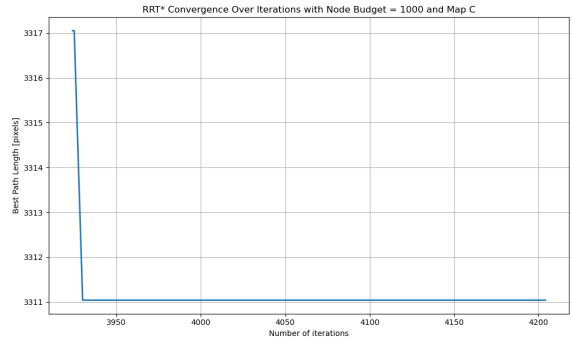


Fig. 64: Convergence of length path along the iterations for Map C, node budget of 1000.

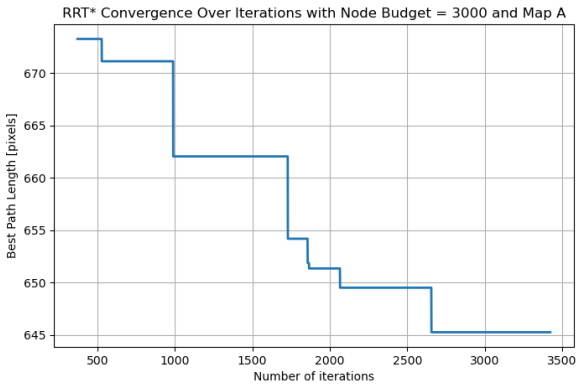


Fig. 65: Convergence of length path along the iterations for Map A, node budget of 3000.

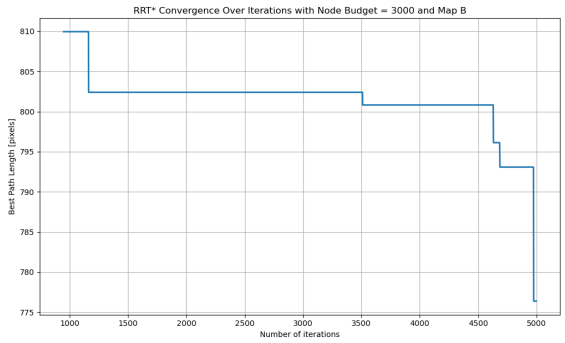


Fig. 66: Convergence of length path along the iterations for Map B, node budget of 3000.

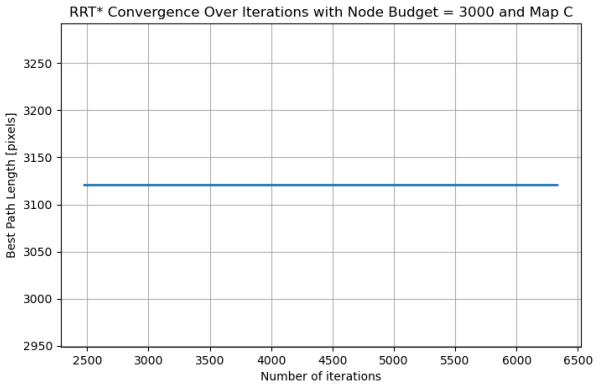


Fig. 67: Convergence of length path along the iterations for Map C, node budget of 3000.

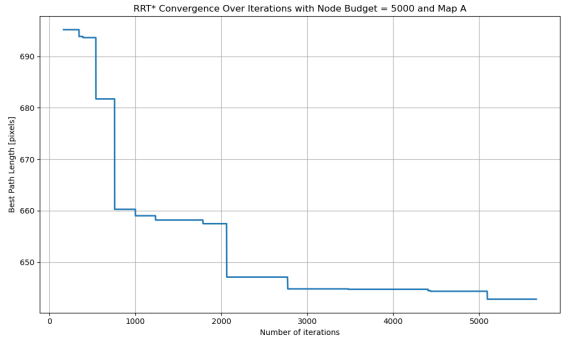


Fig. 68: Convergence of length path along the iterations for Map A, node budget of 5000.

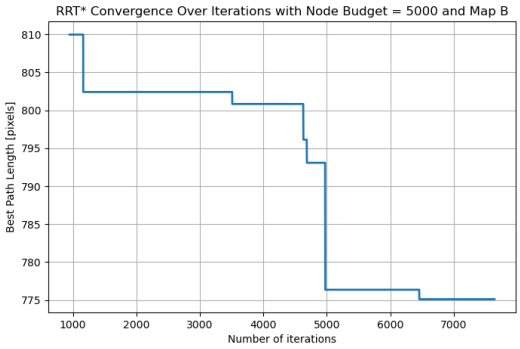


Fig. 69: Convergence of length path along the iterations for Map B, node budget of 5000.

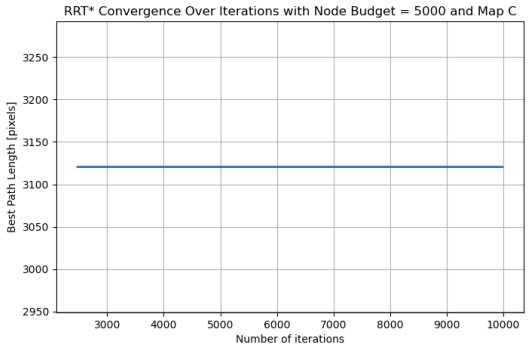


Fig. 70: Convergence of length path along the iterations for Map C, node budget of 5000.

The plots for step size 15 with node budget 5000 are already given in Figures 68-70 .Evolution of the path length along the iterations for other step sizes, with the same node budget of 5000 :

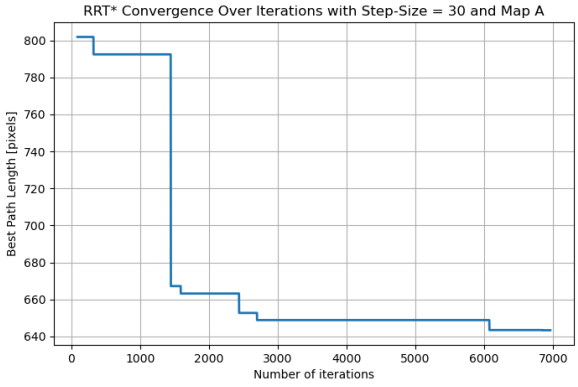


Fig. 71: Convergence of length path along the iterations for Map A, step size of 30.

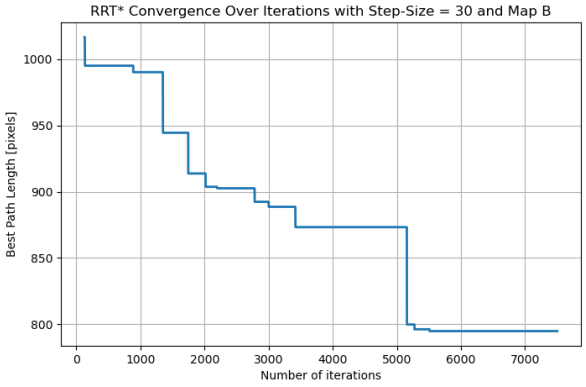


Fig. 72: Convergence of length path along the iterations for Map B, step size of 30.

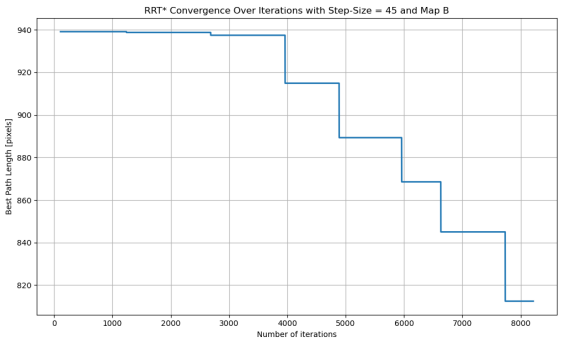


Fig. 75: Convergence of length path along the iterations for Map B, step size of 45.

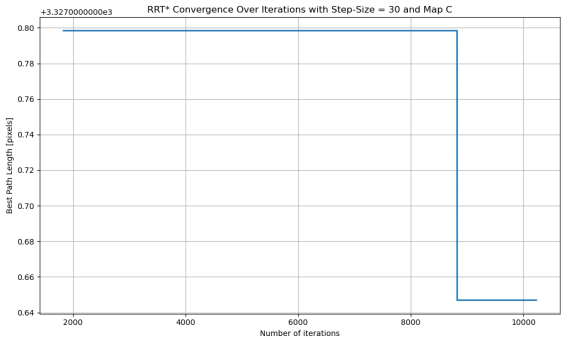


Fig. 73: Convergence of length path along the iterations for Map C, step size of 30.

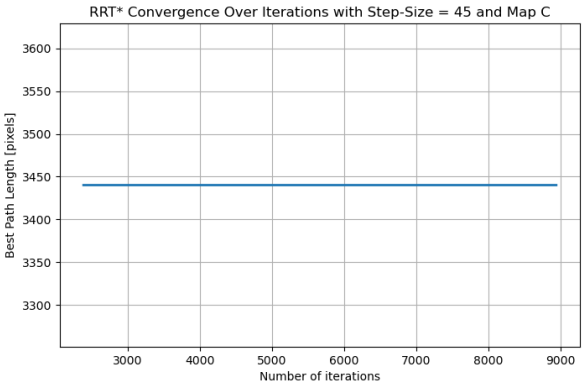


Fig. 76: Convergence of length path along the iterations for Map C, step size of 45.

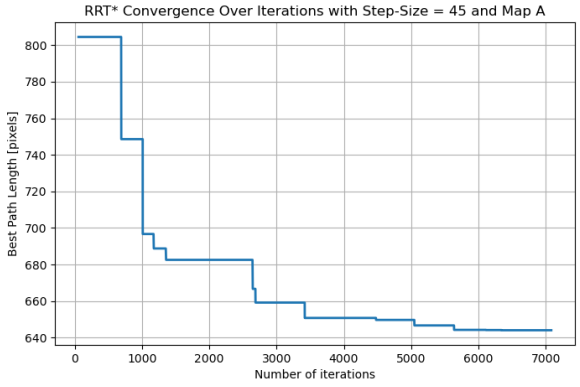


Fig. 74: Convergence of length path along the iterations for Map A, step size of 45.

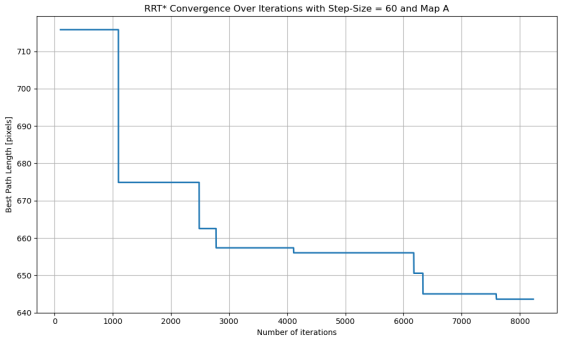


Fig. 77: Convergence of length path along the iterations for Map A, step size of 60.

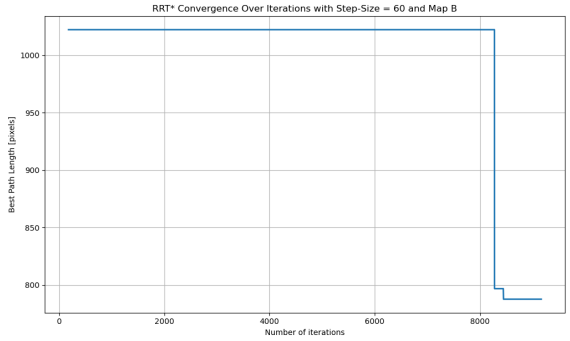


Fig. 78: Convergence of length path along the iterations for Map B, step size of 60.

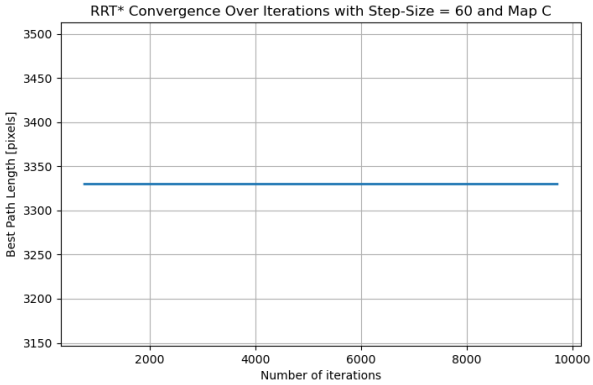


Fig. 79: Convergence of length path along the iterations for Map C, step size of 60.

C. *RRT-connect results*

Paths with different node budget:

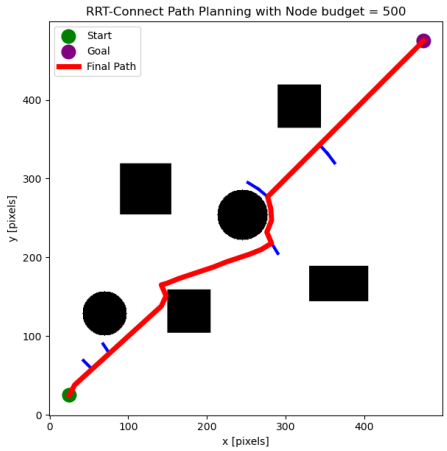


Fig. 80: Map A with RRT-connect with a node budget of 500.

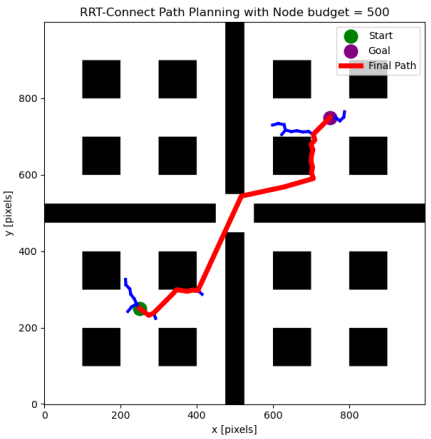


Fig. 81: Map B with RRT-connet with a node budget of 500.

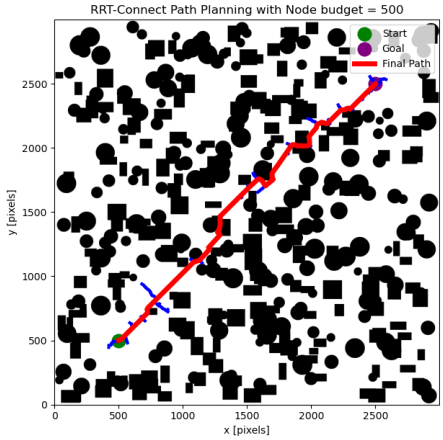


Fig. 82: Map C with RRT-connect with a node budget of 500.

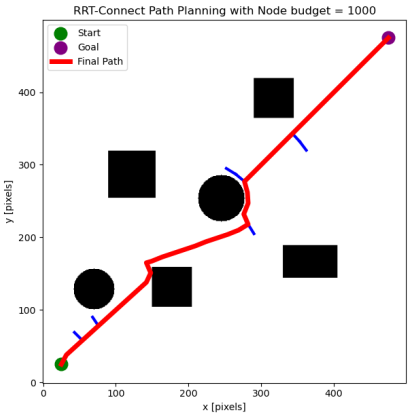


Fig. 83: Map A with RRT-connect with a node budget of 1000.

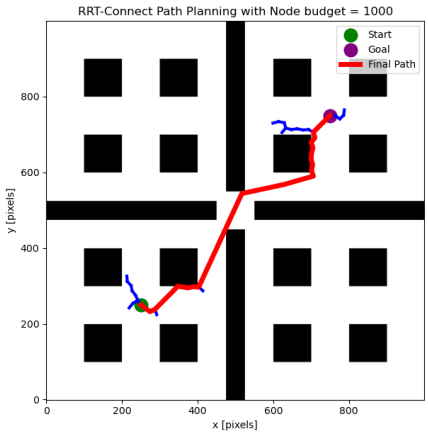


Fig. 84: Map B with RRT-connet with a node budget of 1000.

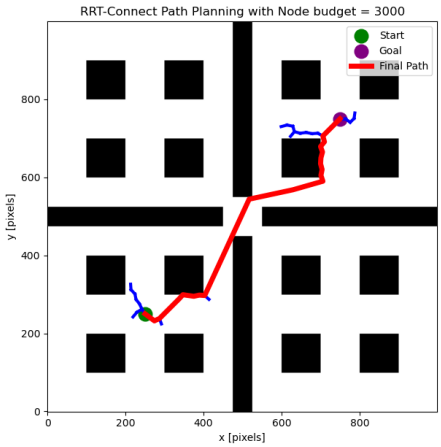


Fig. 87: Map B with RRT-connet with a node budget of 3000.

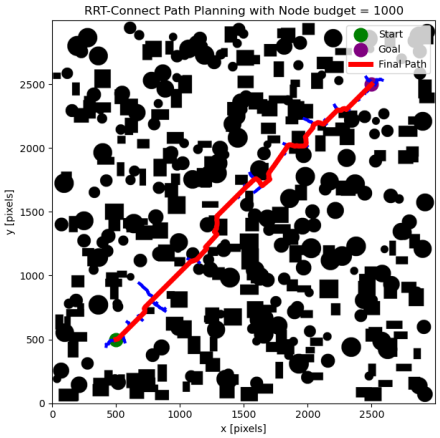


Fig. 85: Map C with RRT-connect with a node budget of 1000.

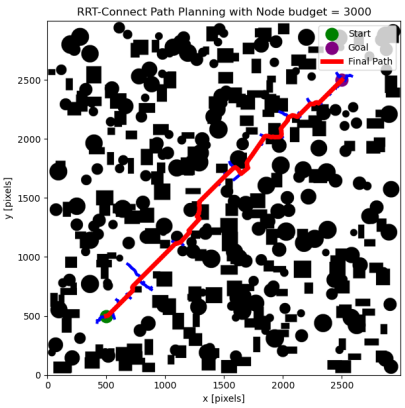


Fig. 88: Map C with RRT-connect with a node budget of 3000.

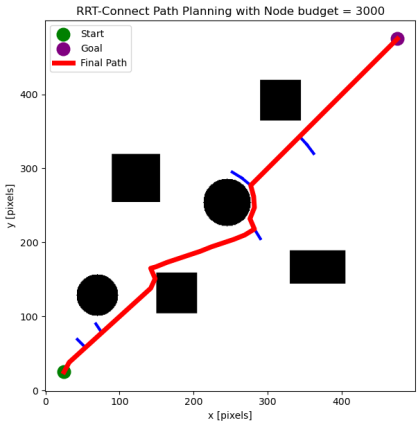


Fig. 86: Map A with RRT-connect with a node budget of 1000.

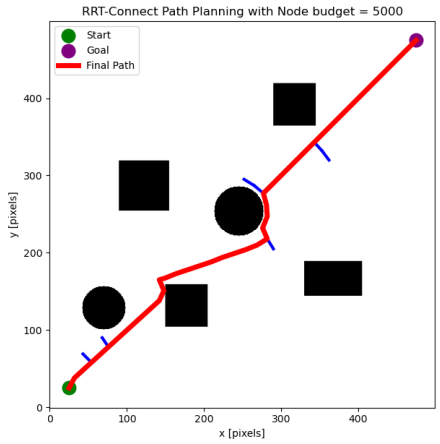


Fig. 89: Map A with RRT-connect with a node budget of 5000.

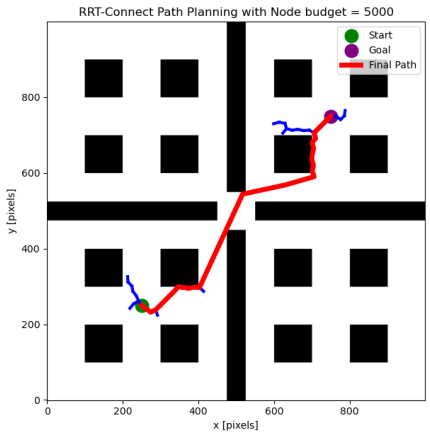


Fig. 90: Map B with RRT-connet with a node budget of 5000.

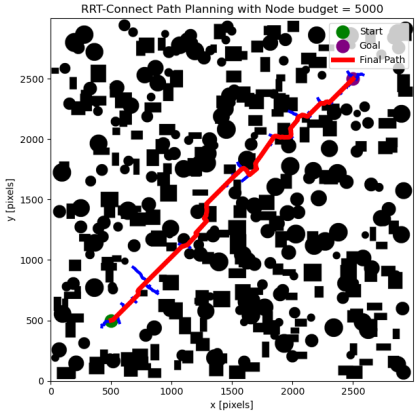


Fig. 91: Map C with RRT-connect with a node budget of 5000.

For the different step-sizes , we consider the ones with step size 15 are already given in the Figures 89-91, so we have:

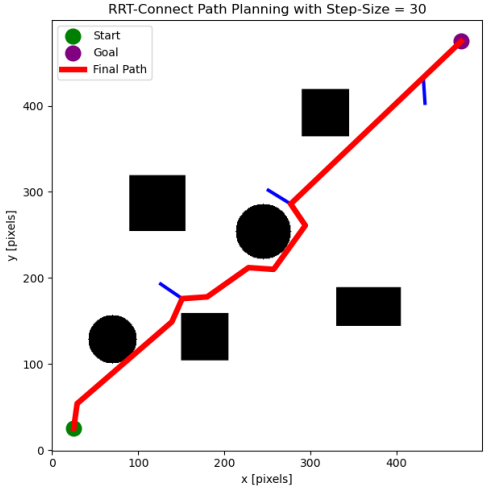


Fig. 92: Map A with RRT-connect with a step size of 30.

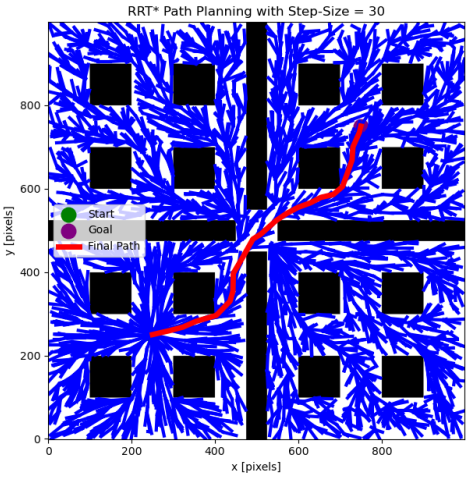


Fig. 93: Map B with RRT-connect with a step size of 30.

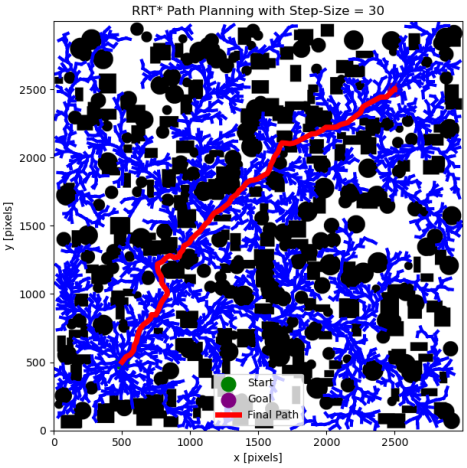


Fig. 94: Map C with RRT-connect with a step size of 30.

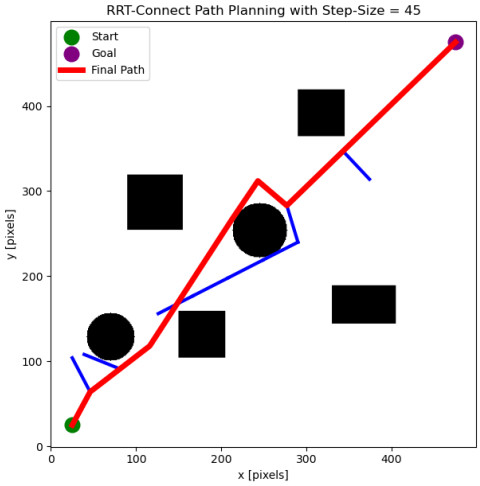


Fig. 95: Map A with RRT-connect with a step size of 45.

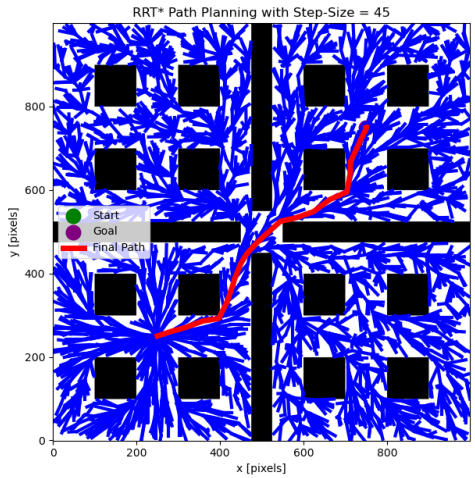


Fig. 96: Map B with RRT-connect with a step size of 45.

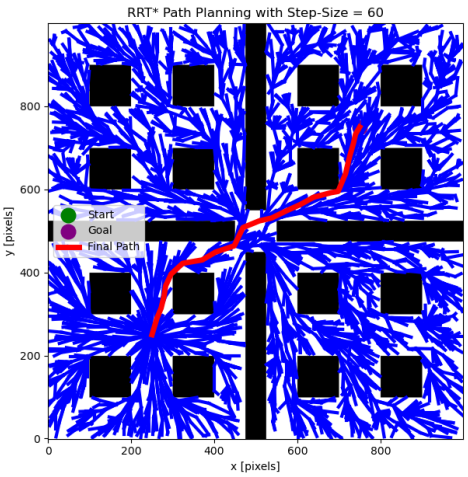


Fig. 99: Map B with RRT-connect with a step size of 60.

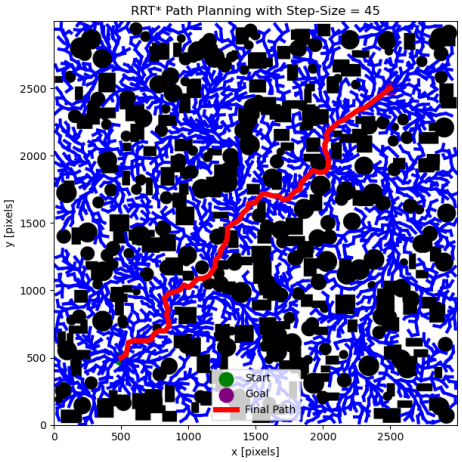


Fig. 97: Map C with RRT-connect with a step size of 45.

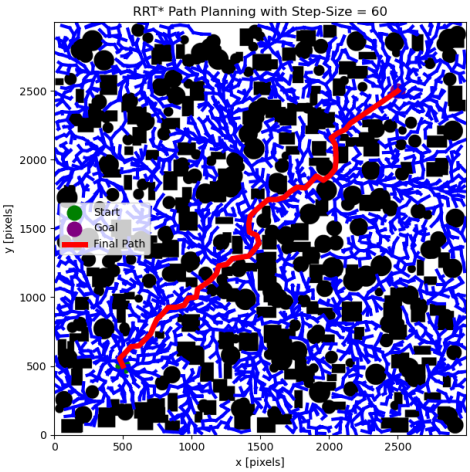


Fig. 100: Map C with RRT-connect with a step size of 60.

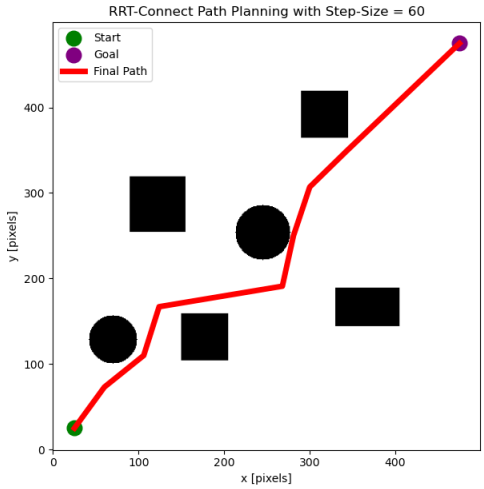


Fig. 98: Map A with RRT-connect with a step size of 60.

D. RRT-adaptive bias results

Paths with different node budgets:

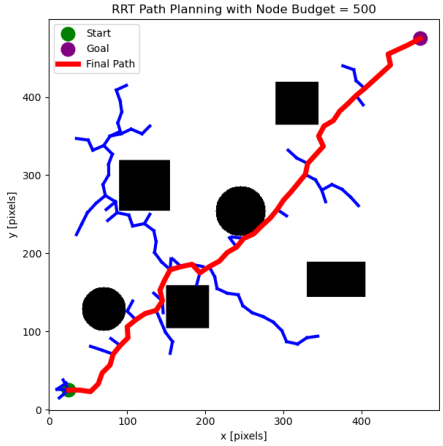


Fig. 101: Map A with RRT-adaptive bias with a node budget of 500.

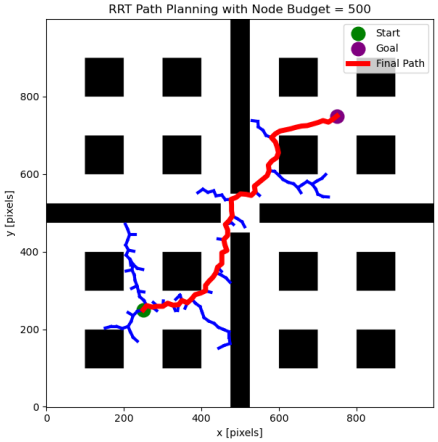


Fig. 102: Map B with RRT- adaptive bias with a node budget of 500.

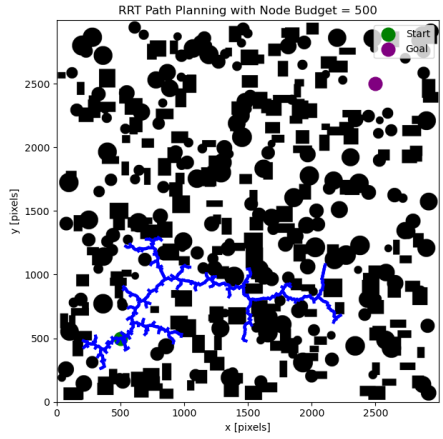


Fig. 103: Map C with RRT-adaptive bias with a node budget of 500.

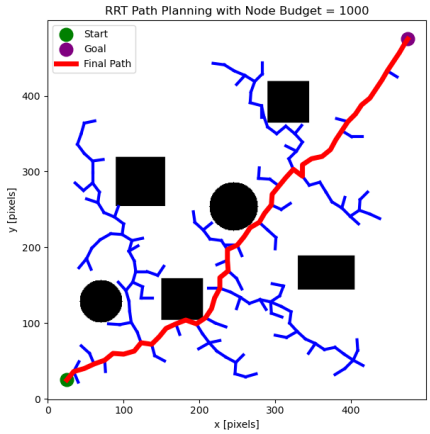


Fig. 104: Map A with RRT-adaptive bias with a node budget of 1000.

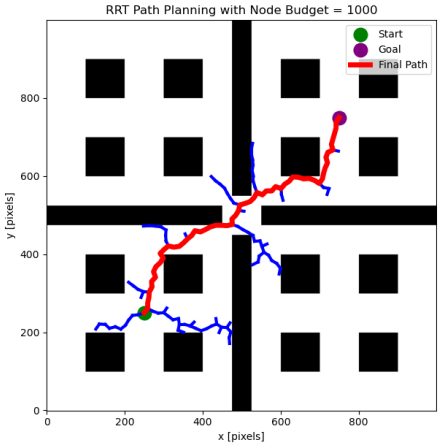


Fig. 105: Map B with RRT- adaptive bias with a node budget of 1000.

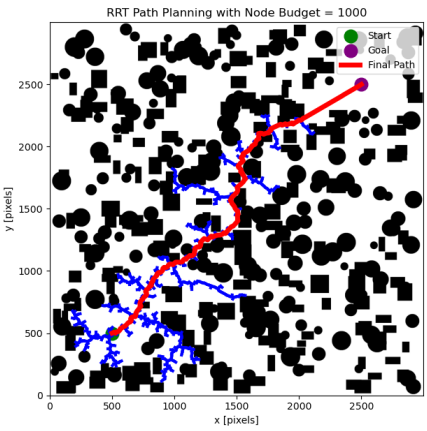


Fig. 106: Map C with RRT-adaptive bias with a node budget of 1000.

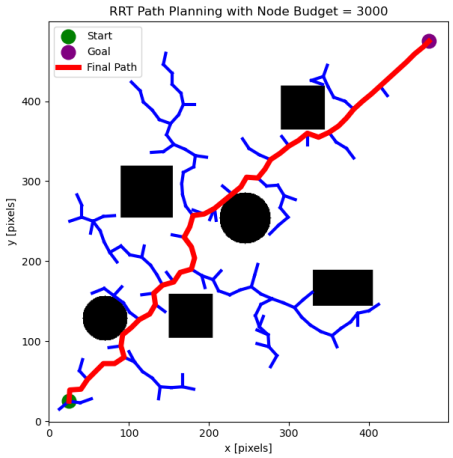


Fig. 107: Map A with RRT-adaptive bias with a node budget of 3000.

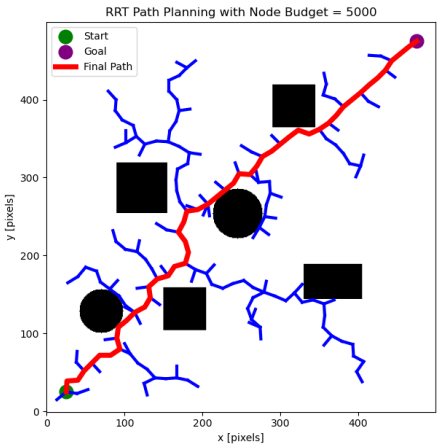


Fig. 110: Map A with RRT-adaptive bias with a node budget of 5000.

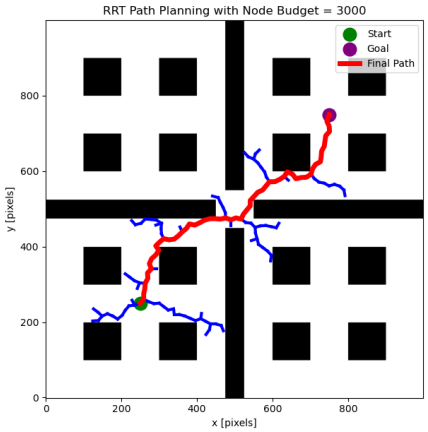


Fig. 108: Map B with RRT- adaptive bias with a node budget of 3000.

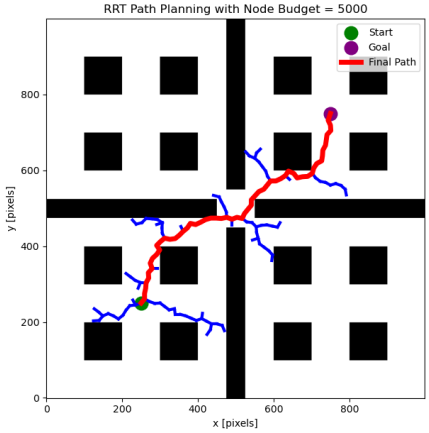


Fig. 111: Map B with RRT- adaptive bias with a node budget of 5000.

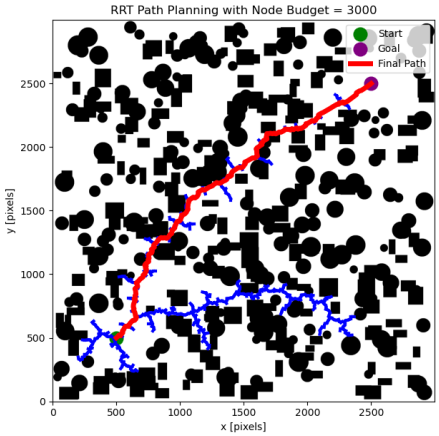


Fig. 109: Map C with RRT-adaptive bias with a node budget of 3000.

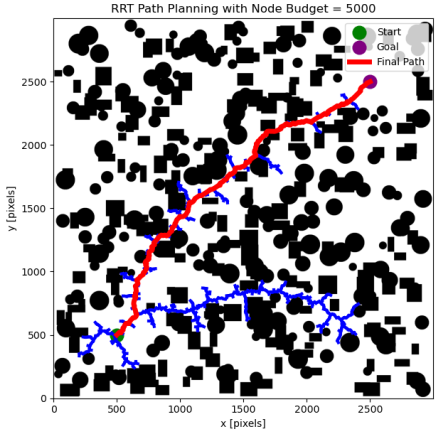


Fig. 112: Map C with RRT-adaptive bias with a node budget of 5000.

For the different step-sizes , we consider the ones with step size 15 are already given in the Figures 110-112, so we have:

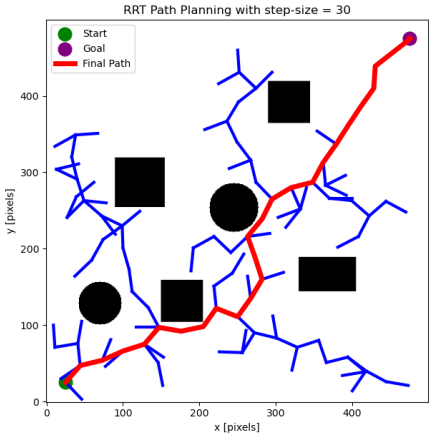


Fig. 113: Map A with RRT-adaptive bias with a step size of 30.

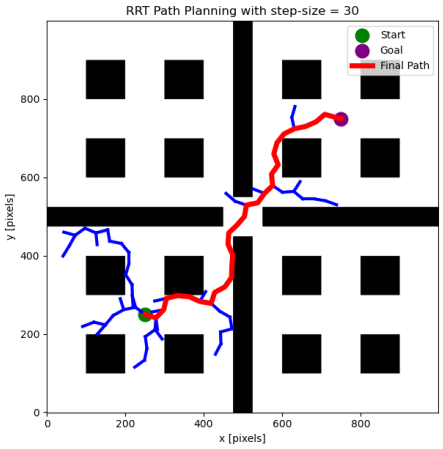


Fig. 114: Map B with RRT-adaptive bias with a step size of 30.

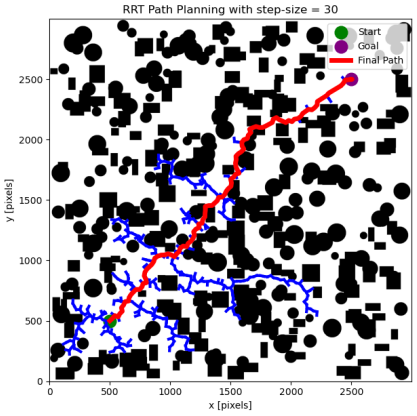


Fig. 115: Map C with RRT-adaptive bias with a step size of 30.

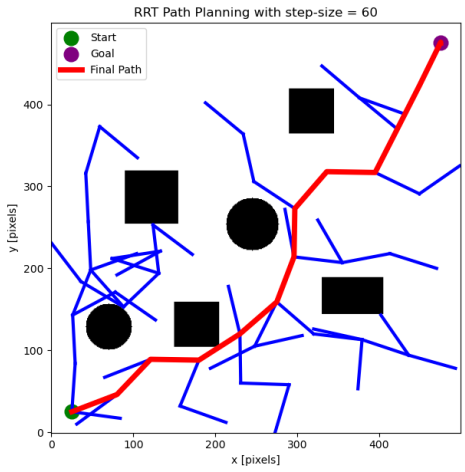


Fig. 116: Map A with RRT-adaptive bias with a step size of 60.

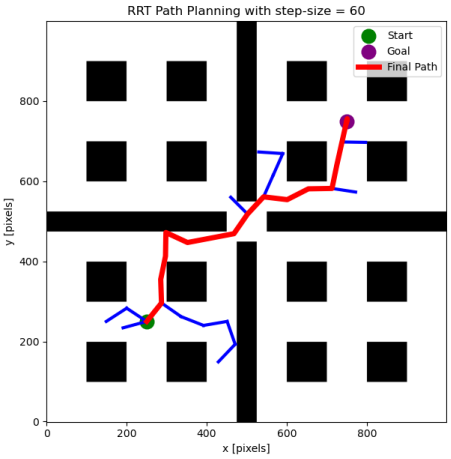


Fig. 117: Map B with RRT-adaptive bias with a step size of 60.

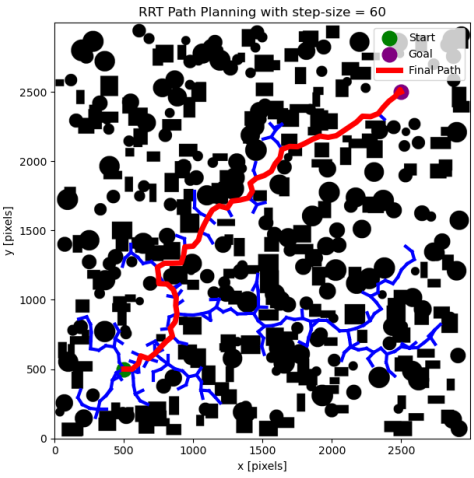


Fig. 118: Map C with RRT-adaptive bias with a step size of 60.

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